

Comprehensive study on vapour-liquid coexistence of the truncated and shifted Lennard-Jones fluid including planar and spherical interface properties

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Abstract

Vapour-liquid equilibria (VLE) of the Lennard-Jones potential, truncated and shifted at 2.5σ are studied with molecular dynamics simulation, which is an attractive potential for studying inhomogeneous systems. Comprehensive simulation data is reported for three cases: no interface, planar interface, or spherical interface between the coexisting phases, covering a wide range of temperatures. Spherical droplets are studied also for a range of radii between 5 and 16σ . The size dependence of surface tension and other properties is quantified for spherical interfaces. All simulation results are correlated with a consistent set of empirical equations. Comparison to other authors as well as to experimental data of noble gases and Methane is given.

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1 Introduction

The thermodynamic description of the vapour-liquid interface goes back to Gibbs [1], who defined it as a physical boundary between two distinct phases, i.e. a region characterized by a local gradient of the density changing from one phase to the other. The first to recognize the smooth variation of density and local pressure tensor across the interface was van der Waals [2].

The vapour-liquid interface has been subject to extensive studies for more than one century because of its importance in many industrial applications, such as phase-change heat transfer, spread wetting, and material processing. The thickness of the interfacial region is in the nanometre range, making experimental studies of such a thin region extremely difficult. Although there are many experimental studies on vapour-liquid interfaces, physical understanding of interfacial phenomena still relies heavily on theoretical analysis and numerical simulations. Molecular dynamics simulation is one of the most effective numerical tools to study interfacial phenomena since it yields detailed information on the molecular structure of an interface on the basis of a molecular model.

Three types of interface geometry are common: planar interfaces (e.g. evaporation processes in the natural convection regime), spherical interfaces (e.g. condensation processes where nanometre scale droplets emerge), and cylindrical interfaces (e.g. free jets). In the present work, planar and spherical interfaces are investigated only, because cylindrical interfaces with nanometre scale diameters seem technically less important. Homogeneous vapour-liquid (VLE) are also determined with high accuracy to provide the thermodynamic properties without any influence of an interface.

A longstanding issue is the size dependence of properties such as surface tension or liquid density of very small droplets [3, 4], which are important, e.g. for nucleation theories.

The present simulative investigation is based on a very simple molecular model, the Lennard-Jones potential, truncated and shifted at 2.5σ . It is one of the most basic and computationally inexpensive molecular models, but it is sufficiently realistic at least for some noble gases. So, it has been used in numerous theoretical studies on coexistence [5–15].

Results cover in detail most of the temperature range between triple point and critical point. Extensive comparison to data of other authors is given, which, however, generally only study a much narrower range of states. To provide tools for subsequent work, empirical correlations of the simulation results are developed. A comparison to experimental data of noble gases and Methane is given as well.

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2 Molecular model

In this work, the truncated and shifted Lennard-Jones potential u_{TS} with a cut-off radius 2.5σ was investigated. It is defined by [16]

$$u_{\text{TS}}(r_{ij}) = \begin{cases} u_{\text{LJ}}(r_{ij}) - u_{\text{LJ}}(2.5\sigma) & r_{ij} \leq 2.5\sigma \\ 0 & r_{ij} > 2.5\sigma, \end{cases} \quad (1)$$

where u_{LJ} is the usual Lennard-Jones potential

$$u_{\text{LJ}}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], \quad (2)$$

σ and ϵ are the size and energy parameters, and r_{ij} is the distance between two molecules i and j . In equation (2), the physically appropriate r^{-6} -term describes dispersive attraction acting at medium distances and the empirical r^{-12} -term describes repulsion acting at short distances. As intermolecular interactions vanish for distances above 2.5σ by definition of this potential, no long range corrections have to be considered. This makes truncated potentials particularly attractive for simulative investigations of fluids where different phases are present. Tedious and disputed long-range corrections that influence the properties of the system do not have to be considered [12, 17, 18].

In molecular thermodynamics, physical properties are conveniently reduced with the potential parameters σ and ϵ as well as the molecular mass m , i.e. temperature $T^* = k_B T / \epsilon$, density $\rho^* = \rho \sigma^3$, pressure $p^* = p \sigma^3 / \epsilon$, enthalpy $h^{\text{res}*} = h^{\text{res}} / \epsilon$, surface tension $\gamma^* = \gamma \sigma^2 / \epsilon$, length $z^* = z / \sigma$, and time $t^* = t \sqrt{\epsilon / m} / \sigma$. These reduced properties are used here throughout.

3 Homogeneous bulk VLE simulations

The Grand Equilibrium method [19] was used to determine the homogeneous bulk VLE of the truncated and shifted Lennard-Jones potential at specified temperatures. A brief outline of that method is given in the following: in the first step, one isobaric-isothermal (NpT) simulation in the liquid phase is performed either by molecular dynamics or Monte-Carlo technique. During this simulation, the chemical potential and its derivative with respect to the pressure, i.e. the molar volume, are calculated. From these results, a first order Taylor series expansion for the chemical potential in the liquid as a function of the pressure $\mu^l(p)$ is obtained. In the second step, one pseudo grand canonical (pseudo- μVT) simulation is performed for the vapour phase. In this vapour phase simulation, using the function $\mu^l(p)$, the chemical potential is specified according to the instantaneous pressure. After equilibration, the dew point is sampled yielding all thermodynamic properties including vapour pressure p_σ . Finally, bubble point properties are obtained from first Taylor series expansions around the liquid simulation results.

For the liquid, molecular dynamics simulations were performed in the NpT ensemble using isokinetic velocity scaling [16] and Anderson's barostat [20]. The number of molecules was 1372 throughout and the reduced time step Δt^* was 0.004. The initial configuration was a face centred cubic lattice, the fluid was equilibrated over 10 000 time steps with the first 2 000 time steps in the canonic (NVT) ensemble. The production run went over 300 000 time steps with a reduced membrane mass Mm/σ^4 of 0.0001. Widom's insertion method [21] was used to calculate the chemical potential inserting 6860 test molecules each time step. The number of molecules for vapour phase simulations was on average 500. After 2 000 initial NVT Monte-Carlo cycles, starting from a face centred cubic lattice, 8 000 equilibration cycles in the pseudo- μVT ensemble were performed. The length of the production run was 100 000 cycles.

VLE were calculated in the temperature range from $T^* = 0.64$ to 1.06 with an increment of $\Delta T^* = 0.03$. Table 1 compiles the results together with their statistical uncertainties which were obtained by block averaging [22] and following the error propagation law.

In order to facilitate the use of these data for subsequent comparison, vapour pressures p_σ^* , bubble densities ρ'^* , dew densities ρ''^* , and enthalpies of vaporization Δh_v^* were correlated. The correlation for the vapour pressure is given by

$$\ln p_\sigma^* = 3.1664 - 5.9809/T^* + 0.01498/T^{*3}, \quad (3)$$

and shown together with present and other simulation results in figure 1. It should be pointed out that this figure and the following also contain data for real fluids which will be discussed in Section 5. The accuracy of this fit can better be seen in a deviation plot, cf. figure 2 a), where vapour pressure correlation and our simulation data typically agree within 1%. The relative deviation of a property y is defined here throughout as $\delta y = (y_{\text{sim}} - y_{\text{cor}})/y_{\text{cor}}$. Comparison with the results of Mareschal et al. [5] and Adams and Henderson [6] shows an agreement within 3%. The Monte-Carlo results of Trokhymchuk and Alejandre [12] are represented typically within 4%, while their molecular dynamics results show deviations of only 1%. Data by Camp and Allen [9], for which the authors claim error bars of about 5%, shows the largest deviations of up to 37% at lower temperatures.

To correlate the saturated densities, Guggenheim type equations are used [23]

$$\rho'^* = \rho_c^* + 0.5649 \cdot (T_c^* - T^*)^{1/3} + 0.1314 \cdot (T_c^* - T^*) + 0.0413 \cdot (T_c^* - T^*)^{3/2}, \quad (4)$$

$$\rho''^* = \rho_c^* - 0.5649 \cdot (T_c^* - T^*)^{1/3} + 0.2128 \cdot (T_c^* - T^*) + 0.0702 \cdot (T_c^* - T^*)^{3/2}. \quad (5)$$

From this, the critical temperature and density are found to be $T_c^* = 1.0779$ and $\rho_c^* = 0.3190$, respectively. These data agree very well both with values of Shi and Johnson [7] who reported $T_c^* = 1.0795$ and $\rho_c^* = 0.3211$ and also with Smit [8] who reported $T_c^* = 1.085$ and $\rho_c^* = 0.317$. Figure 3 depicts the present saturated density simulation results with the corresponding correlations, cf. equations (4) and (5), and the results of other authors. Data by Shi and Johnson [7] are confined to a range of elevated temperatures above $T^* = 0.95$, while Camp and Allen [9] covered a larger range of intermediate temperatures.

The deviation plot in figure 2 b) shows that our bubble densities are represented by the correlation typically within 0.1%, except close to the critical point. The results of Shi and Johnson [7] show deviations from -0.1% at $T^* = 0.95$ up to 2% at $T^* = 1.04$. All results of Camp and Allen [9] agree within their large error bars of around 3% .

Present dew densities, cf. figure 2 c), show a scatter of typically 2% , whereas the corresponding results of Shi and Johnson [7] show systematic deviations ranging from 1.8% at $T^* = 0.95$ up to -6% at $T^* = 1.04$. Results of Camp and Allen [9] vary typically within 20% .

Finally, the enthalpy of vaporization was also correlated

$$\Delta h_v^* = 6.8285 \cdot (1 - T^*/T_c^*)^{0.1889 - 0.1395 \cdot (T^*/T_c^*) + 0.2727 \cdot (T^*/T_c^*)^2}. \quad (6)$$

Figure 4 depicts the correlation and available simulation results. The quality of this fit can also better be seen in a deviation plot, cf. figure 2 d), where present simulation data and correlation agree typically within 0.2%. The corresponding data of Camp and Allen [9] are systematically lower by about 2%, which is in the same order as their uncertainty.

It can be summarized that the present homogeneous bulk VLE simulations results show low statistical uncertainties, cover most of the temperature range from triple point to critical point and agree well with data of other authors [5–7, 9, 12].

4 Planar interface VLE simulations

The most straightforward way to simulate vapour-liquid equilibria mimics the experimental setup. Two coexisting phases in direct contact separated by an interface are simulated in a single volume. If periodic boundary conditions are applied in all directions and an appropriate initial configuration is specified, one liquid and one vapour phase, separated by two planar interfaces, will be present. Such planar interface VLE simulations, as one may name them, have been performed by many authors, e.g. Rowlinson and Widom [2], Haye and Bruin [10], and Nijmeijer et al. [13]. An advantage of a direct interfacial simulation is that it provides beside the usual bulk VLE properties information on the interface. However, it has been found that these simulations require a relatively large number of molecules and produce unreliable results near the critical point [24].

To investigate the planar interface, molecular dynamics simulations were performed here in an elongated cuboid shaped volume with periodic boundary conditions in all directions. The volume had a length of l in x and y direction and a length of $6l$ in z direction. It contained a vapour and a liquid phase forming two interfaces in the x,y -plane of the simulation volume. The liquid phase was chosen to have a thickness of approximately $2l$, so that the vapour phase had a thickness of $4l$. From this arrangement follows that two vapour-liquid interfaces with an area of $A = l^2$ were simulated. System size was chosen so that the liquid contained approximately 2700 molecules leading to an l of about 12.5σ . So the liquid slab had a thickness of about 25σ , which is tenfold larger than the maximum range of the investigated potential. This relatively large thickness should ensure that the structure in the centre of the liquid slab is not influenced by the interfaces.

As a preparatory step for obtaining an adequate initial configuration, a homogeneous liquid and a homogeneous vapour phase were simulated independently in a cubic volume at saturated densities taken from the homogeneous VLE simulations, cf. equations (4) and (5). Both simulations were done by molecular dynamics in the NVT ensemble starting from a face centre cubic lattice for 5 000 time steps of $\Delta t^* = 0.004$ to obtain equilibrated vapour and liquid phases. The number of molecules in the liquid was 1372 throughout. The molecular configurations of these equilibrated homogenous phases were then used to construct an initial configuration for the inhomogeneous system by duplicating the liquid configuration and adding vapour molecules in the remaining vacuum. Overlaps were avoided and relative positions as well as velocities were conserved within each phase. With this carefully constructed initial configuration, the simulation of the inhomogeneous system is mechanically stable from the start of the simulation. Only the inhomogeneous systems' centre of mass in z direction was kept in the origin of the z axis

by a proportional controller with a long time constant. Though this is not necessary for surface tension calculations, it is useful for density profile calculations.

Planar interface VLE calculations were carried out in the temperature range from $T^* = 0.625$ to 1.05 with an increment of $\Delta T^* = 0.025$. Therefore, isokinetic NVT ensemble simulations were performed with 200 000 equilibration time steps, followed by a production phase of 1 000 000 time steps. In order to determine the density profile, the cuboid simulation volume was divided into 500 subvolumes perpendicular to the interface (in z direction) and the density profile was obtained by usual time averaging.

To facilitate the use of the density profile results obtained from simulation, an algebraic representation was required. Generally, it is written by

$$\rho(z) = \frac{1}{2}(\rho_l - \rho_v) + \frac{1}{2}(\rho_l + \rho_v)f(z), \quad (7)$$

where ρ_l^* and ρ_v^* are the densities of the liquid and the vapour phase, respectively, far away from the interface, and f is a monotonic function so that $f(\pm\infty) = \pm 1$. Several expressions for $f(z)$ have been proposed in the literature:

- (i) Classical [25]: this profile results from the original van der Waals theory when a classical mean-field equation of state is used

$$f(z) = \tanh\left(\frac{2z}{D}\right), \quad (8)$$

where D is the interfacial thickness.

- (ii) Fisk-Widom [26]: this profile results from replacing the classical equation of state in the original van der Waals theory by a non-classical equation of state

$$f(z) = \frac{\sqrt{2} \tanh(2z/D)}{\sqrt{3 - \tanh^2(2z/D)}}. \quad (9)$$

- (iii) Error-function [27]: this profile is usually associated with the capillary wave theory of Buff et al. [27]. As reported there, capillary wave theory yields a second moment of the density profile which is consistent with an error function profile

$$f(z) = \operatorname{erf}\left(\frac{\sqrt{\pi}z}{D}\right). \quad (10)$$

(iv) Exponential [28]: this profile was proposed by Lekner and Henderson [28] who specified $D = (2 \ln 5)^{-1}t$, where t is the "10-90 thickness" to be taken from molecular simulation [29]

$$\rho(z) = \begin{cases} \rho_l - \frac{1}{2}(\rho_l - \rho_v) \cdot \exp(z/D) & z < 0 \\ \rho_v + \frac{1}{2}(\rho_l - \rho_v) \cdot \exp(-z/D) & z > 0. \end{cases} \quad (11)$$

In the present work, D was used as a free parameter, allowing for greater flexibility.

Other expressions have also been proposed, e.g. the Robert-Stuart profile [30], but were not used here.

Figure 5 depicts a comparison of the four profiles (8) to (11) and simulation data to which they were fitted at $T^* = 0.8$. As simulation results are noisy, the classical profile, cf. equation (8), was chosen as the baseline instead. It can be seen that the classical profile matches simulation data well with slight systematic sinusoidal deviations on the right, i.e. vapour side ($z > 0$) of -0.006 in absolute terms of ρ^* . On the liquid side, simulation data have a larger absolute scatter, but it can be assumed that deviations are roughly in the same order as on vapour side. Somewhat larger deviations of 0.013 are found on the vapour side for the Error-function profile, cf. equation (10), where it is also overestimating the density. The opposite behaviour can be seen for both Fisk-Widom profile, cf. equation (9), and Exponential profile, cf. equation (11), which deviate up to more than 0.03 . Therefore, it is concluded that the classical profile is the most appropriate approximation of those four to describe the density gradient through the interface.

The classical profile, cf. equations (7) and (8), was chosen to fit the whole simulated density profile data set. Both vapour ρ_v^* and liquid ρ_l^* densities far away from the interface as well as the interfacial thickness D^* were obtained in this way; values can be found in table 2. It is interesting to compare the simulated density in the liquid from planar interface VLE calculations with bubble densities from homogeneous VLE calculations in more detail. Figure 6 a) depicts a deviation plot to examine this issue. The baseline of this plot is the bubble density correlation, cf. equation (4), which is assumed to be correct within 0.1% . It can be seen that present simulation data agree within 0.15% for all data points up to $T^* = 0.98$. For temperatures above, the agreement is worse by up to more than an order of magnitude. This finding proves the consistency of the homogeneous and planar interface VLE data sets and shows that bubble densities can be determined accurately by planar interface VLE simulations at temperatures below approximately 90% of T_c . Comparison with the planar interface VLE results of other authors can also be seen in figure 6 a). Mareschal et al. [5], Haye and Bruin [10], and Dunikov et al. [11] are in good agreement with maximum deviations of 0.35% only. Monte-Carlo and molecular dynamics results of Trokhymchuk and Alejandre [12] agree

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typically within 0.5%.

An analogous comparison was made for the vapour density, cf. figure 6 b), where the correlation of the homogeneous data is assumed to be correct within 2%. At temperatures from 0.7 to 0.98 the deviations are practically within 5%, which again supports consistency of the data sets. Above $T^* = 1.0$ deviations increase considerably showing simulative difficulties due to fluctuations at high temperatures. Also at low temperatures larger deviations occur, which are attributed to poor statistics due to the very small number of vapour molecules which drops down to only around 16 in the simulation at $T^* = 0.625$. As shown in figure 6 b), the reported results of Mareschal et al. [5], Haye and Bruin [10], Dunikov et al. [11], and Trokhymchuk and Alejandre [12] agree typically within 5%.

As saturated densities from homogeneous VLE calculations can be assumed to be more accurate, they are preferred here for a general representation following equations (7) and (8) of the simulated planar interface density profiles. So ρ'^* and ρ''^* , cf. equations (4) and (5), were specified for ρ_l^* and ρ_v^* in the next step. Only one free temperature dependent parameter was adjusted to the entire planar interface density profile data set, i.e. the reduced interface thickness D^* . The fitted correlation is shown in figure 7 and given by

$$D^* = 1.720 \cdot \left(\frac{T_c^* - T^*}{T_c^*} \right)^{1.89} + 1.103 \cdot \left(\frac{T_c^* - T^*}{T_c^*} \right)^{-0.62}. \quad (12)$$

The empirically found exponent -0.62 is dominant at high temperatures and close to the reported "universal" exponent -0.63 of Beysens [31]. Correlation (12) and the individual simulation data agree typically within 3%, except for the highest temperature $T^* = 1.05$ with a deviation of 23%. Figure 8 shows a comparison of the general density profile fit and direct simulation data at $T^* = 0.65, 0.85$, and 1.025 . The fit agrees almost perfectly with the simulation results at the lower two temperatures. Slight discrepancies are observed at $T^* = 1.025$. There, the liquid density from planar interface VLE calculation is 2% lower than homogeneous VLE data.

To calculate the surface tension of the planar interface, the virial route was chosen due to its simplicity and reliability [1] while being computationally cheap. The virial route uses the common definition of the surface tension

$$\gamma = \left(\frac{\partial F}{\partial A} \right)_{NVT}. \quad (13)$$

Here, F is the sum of the Helmholtz free energies of the two phases and interfaces. It requires that the interface area is known, so that it is not convenient for surfaces with high curvatures. The usual expression for γ can be derived

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straightforwardly from the fundamental relation between F and the partition function. This is done, e.g. in [2, 17]

for a surface in the x, y -plane and leads to

$$\gamma = \frac{1}{2A} \left\langle \sum_{i=1}^N \sum_{j>i}^N \left(1 - \frac{3z_{ij}^2}{r_{ij}^2} \right) r_{ij} \frac{\partial u(r_{ij})}{\partial r_{ij}} \right\rangle, \quad (14)$$

which includes the virial $r_{ij} \partial u(r_{ij}) / \partial r_{ij}$.

Surface tension results are given in table 2 and shown in figure 9. It is evident that the surface tension vanishes at the critical temperature. The trend of the present simulation results reflects this. Simulation data were correlated by

$$\gamma^* = 2.08 \cdot (1 - T^*/T_c^*)^{1.21}, \quad (15)$$

where the empirically found exponent 1.21 is close to the value of the "critical index" 1.26 from fluctuation theory of critical phenomena [32]. The reliability of this fit can better be seen in the corresponding deviation plot, cf. figure 10, where the scatter is typically 2%, except close to the critical temperature at $T^* = 1.05$ with a deviation of 18%. There is a systematic deviation between present surface tension results and those of Haye and Bruin [10] which are found to be lower by about 5%. But the results of both Chen [14] and Holcomb et al. [15] at $T^* = 0.72$ match very well with the present data with differences of only 0.02% and 0.4%, respectively. Also a good agreement with the results of Mareschal et al. [5] is observed where the deviations are within 3%. The deviations of the Monte-Carlo and molecular dynamics results of Trokhymchuk and Alejandre [12] are typically within 3% as well. The results of Dunikov et al. [11] match within 1%, except at $T^* = 0.926$. Results of Nijmeijer et al. [13] agree within 4%, except at $T^* = 0.92$ with a deviation of 18%.

From present planar interface simulation results, using surface tension γ and characteristic length of the interface area l , also the mean square amplitude of capillary waves can be found [33, 34].

In summary, a planar interface VLE data set is presented together with correlations being in good agreement with other authors [5, 10–12] and corresponding well to homogeneous VLE results.

5 Real substances

The use of the Lennard-Jones potential to describe the intermolecular interactions of Argon in computer simulation goes back to the early nineteen sixties [35, 36]. Thereafter many new parameter sets have been reported, e.g. in [37–39]. In order to prove the appropriateness of the truncated and shifted Lennard-Jones potential here regarding thermodynamic properties, the parameters σ and ϵ were adjusted to experimental vapour pressure and bubble density from equations of state for Argon [40], Krypton [41], Xenon [41], and Methane [42] using correlations (3) and (4). The noble gases were chosen as they are rotationally symmetric and exclusively interact through repulsion and dispersive attraction. Methane molecules are very compact so that it is reasonable to assume rotational symmetry. Furthermore, the weak octupolar interactions between Methane molecules, which decay fast with r^{-7} , can be neglected. The parameters resulting from the fit are given in table 3.

Figures 1, 2 a), b), and 3 show the descriptive quality the truncated and shifted Lennard-Jones potential for all four substances. For convenience, real substance data is shown throughout in reduced units. By examining the absolute values in figures 1 and 3, hardly any discrepancy can be discerned, except for the critical region. Therefore, the deviation plots have to be regarded. The vapour pressure, cf. figure 2 a), can be described mostly within 2%, which is just twofold of the underlying fit accuracy. In figure 2 b) it is found that bubble densities can be described within 0.1% which is underlying fit accuracy, except above 93% of T_c^* of the model. These increasing positive systematic deviations are explained by the fact that all real substances have higher reduced critical temperatures by approximately 1.5%.

Figures 2 c), d), 4, 9, and 10 examine other properties, not used for potential parameter adjustment. Dew densities, cf. figure 2 c), are predicted within 2%, except above 93% of T_c^* , which is also not significant with respect to the accuracy of the underlying fit. Again, increasing negative deviations for high temperatures are explained by different critical temperatures. It is not surprising that the dew densities show the same magnitude of deviations as the vapour pressures, as both are closely related. In figures 2 d) and 4 the predictions of enthalpies of vaporization are shown. Below 93% of T_c^* real substance data is predicted within 2%, where for most of the temperature range negative deviations are observed. This is also not surprising as enthalpy of vaporization, vapour pressure and saturated densities are related by the Clausius-Clapeyron equation [43]. Again, increasing deviations in the critical region are caused by differing critical temperatures.

The predictions of the surface tension for Argon, Krypton, Xenon, and Methane are shown in figures 9 and 10. It

can be seen that the agreement between the model fluid and the noble gases is good, 6% for Argon and Xenon as well as 10% for Krypton in most of the temperature range. But for Methane, larger systematic deviations are found which are attributed to the neglect of the molecular properties discussed above.

At least for the noble gases it can be concluded that simulation with the investigated potential model yield a reliable description of both VLE and interface properties. From this it can be speculated that the simulations of nanometre scale droplets in the following chapter, where comparison to the experiment is not possible, provide even quantitatively correct results for these substances.

6 Spherical interface VLE simulations

Properties of coexisting phases separated by curved interfaces are affected by the curvature. Whereas in equilibrium, the temperature T is the same in the two phases and throughout the interface, it is well known that the pressure p_l inside liquid droplets is higher than in the surrounding vapour p_v , as Laplace equation suggests (see below). It should be noted that the vapour phase in equilibrium with a liquid droplet must have a pressure p_v higher than vapour pressure $p_\sigma(T)$ of the fluid without curved interface. Hence, it corresponds to a state for which the homogeneous fluid is supersaturated. This is in line with the finding that droplets of finite size can only be formed from supersaturated vapours.

It follows from the brief discussion above that not only the the pressure inside p_l and outside the droplet p_v are higher than $p_\sigma(T)$, but also the corresponding densities ρ_l and ρ_v are higher than the saturated densities of the homogeneous fluid $\rho'(T)$ and $\rho''(T)$, respectively. Furthermore, the surface tension decreases with increasing curvature. These effects are quantified in the present work for the truncated and shifted Lennard-Jones potential.

In analogy to the planar interface investigation, the first step to simulate the spherical interface lies in the preparation of a liquid droplet in equilibrium with its vapour. Therefore, molecular dynamics NVT simulations of the liquid were performed for 32 000 particles in a cubic volume at corresponding bubble densities, cf. equation (4). For the vapour, configurations at corresponding dew densities, cf. equation (5), were obtained for 864 particles. The vapour box length was $L_v^* = 42.78$ to 57.28 and each of these runs was performed over 5 000 time steps.

Next, a spherical droplet was created by taking the molecules contained in a given sphere within the liquid cube. All molecules within a spherical volume of same size the centre of the vapour cube were removed, leaving the remaining "vapour" particles untouched. The liquid and vapour were then merged in the vapour volume, conserving relative positions and velocities within each phase. To realize different levels of curvature, spherical liquid droplets of different size surrounded by vapour in equilibrium were simulated subsequently. For all purposes of the present work, the coordinate system originates in the centre of mass of the droplet. To ensure this in the simulations, a proportional controller with a long time constant was used.

Spherical interface VLE calculations were performed at temperatures ranging from $T^* = 0.65$ to 0.95 with an increment of $\Delta T^* = 0.05$. All together 88 droplets of different size were simulated in this temperature range. Most simulations were equilibrated for 10^6 time steps, followed by 10^6 production time steps, whose reduced value was 0.004. For some of the very small droplets up to $3 \cdot 10^6$ production time steps were used to improve statistics.

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The long equilibration time was necessary to reduce the effect of surface and size oscillations of the droplet on the averaged results. In every production time step the local density and the Irving-Kirkwood normal pressure were calculated. The shell thickness for this purpose was $\delta r^* = 0.03$ and the maximum radius was half the length of the cubic volume. Since the liquid phase is always exchanging molecules with the vapour, the droplet shows size oscillations in the time evolution of the system. Especially in simulations of very small droplets, size oscillations of molecules cause quite large changes of the droplet size, yielding significant scatter. Statistics are generally worse in the centre of the droplet due to the small shell volume for averaging.

Some typical density profile simulation results are shown in figure 11. To represent the density profiles, analytic functions were fitted to the simulation data using an expression following Baus et al. [5, 44]

$$\rho = \frac{1}{2}(\rho_l + \rho_v) - \frac{1}{2}(\rho_l - \rho_0) \tanh\left(\frac{2}{D_{\rho l}}(r - r_{\rho l})\right) + \frac{1}{2}(\rho_v - \rho_0) \tanh\left(\frac{2}{D_{\rho v}}(r - r_{\rho v})\right), \quad (16)$$

where ρ_l, ρ_v and ρ_0 characterize the density in the bulk liquid, the bulk vapour and the density minimum, respectively, and r is the radial coordinate. Furthermore, $r_{\rho l}$ and $r_{\rho v}$ denote the radii where the density profile has inflection points on the liquid and vapour side of the interface. In analogy, $D_{\rho l}$ and $D_{\rho v}$ characterize the width of the interface region on the liquid and the vapour side of the interface, respectively. In the fitting procedure of equation (16), the density profile simulation data is weighted by its distance from the centre of the droplet.

To characterize spherical interfaces geometrically on the basis of the density profile, the Gibbs absorption dividing surface R_e [45, 46] has been proposed. It is a radius and given for a spherical droplet centred in the origin of the coordinate system by [46]

$$R_e^3 = \frac{\int dr r^3 \rho}{\int dr \rho}, \quad (17)$$

which can be calculated using equation (16).

Our approach was to first fit equation (16) individually to all 88 droplet simulation data sets and determine their R_e through equation (17). All parameters can be found in table 4 where also the average number of molecules inside a droplet N_L is given which was obtained following a neighbour scheme [47–49].

In the next step, a generalized density profile was derived, being both temperature and size dependent $\rho(T, R_e)$, using the full data set for an overall fit. The densities in the centre of the liquid droplet and in the surrounding

vapour far away from the interface, considering both temperature and size dependence, were correlated using

$$\rho_l^* = \rho'^* + \frac{0.1485 - 0.0471 \cdot (T_c^* - T^*)^2}{R_e^* + 1.272 - 0.4944 \cdot (T_c^* - T^*)^{-1} + 5.0090 \cdot (T_c^* - T^*)}, \quad (18)$$

and

$$\rho_v^* = \rho''^* + \frac{-0.0541 + 0.1682 \cdot T^{*2}}{R_e^* - 1.0810 - 0.2490 \cdot (T_c^* - T^*)^{-1} + 10.87 \cdot (T_c^* - T^*)}, \quad (19)$$

where ρ'^* and ρ''^* are the saturated densities from homogeneous VLE simulation results, which were calculated from equations (4) and (5). For $R_e^* \rightarrow \infty$, ρ_l^* and ρ_v^* are identical with ρ'^* and ρ''^* .

Furthermore, for a generalized correlation, based on equation (16), the following functional form was used for

ρ_0^* , $r_{\rho_l}^*$, $r_{\rho_v}^*$, $D_{\rho_l}^*$, and $D_{\rho_v}^*$

$$g = a_1 + a_2 \cdot (T_c^* - T^*) + a_3 \cdot (T_c^* - T^*)^{-1} + [a_4 + a_5 \cdot T^* + a_6 \cdot (T_c^* - T^*)^{-1}] \cdot R_e^*, \quad (20)$$

where g denotes any of those five properties. The parameters are given in table 5. It is reasonable that the four properties ($r_{\rho_l}^*$, $r_{\rho_v}^*$, $D_{\rho_l}^*$, $D_{\rho_v}^*$), which represent geometric features of the interface, depend linearly on the Gibbs absorption dividing surface, cf. equation (20). The density parameter ρ_0^* is used to describe the weak density minimum, cf. figure 12 (not discernable in figure 11). Simulation data there was the noisiest data to be fitted and therefore the simple correlation (20) was also used for ρ_0^* .

Exemplarily, simulation results and generalized density profile correlation, i.e. equations (16) and (18) to (20), are shown for $T^* = 0.8$ in figure 11. The fit is good enough that it is hardly discernable from the simulation data. Figure 12 shows the results of the fit for $R_e^* = 7, 11$, and 16 at $T^* = 0.65$ and 0.90 . It can be seen that the liquid density inside the droplet is increased by 2% for $T^* = 0.65$ and 3.5% for $T^* = 0.90$ for small droplets of $R_e^* = 7$, containing roughly 500 particles. Also, the vapour density is increased due to different equilibrium conditions.

Irving and Kirkwood [50] reported a thorough definition of the pressure tensor for spherical interfaces. Other formulations, e.g. Harasima pressure tensor, are also described in the literature [2,51]. Due to the intrinsic properties of the Irving-Kirkwood method its statistics are superior [52] and this method is preferred here. Detailed information on the Irving-Kirkwood pressure tensor for a spherical interface can be found in [2,45]. To calculate the pressure tensor for such geometries it is sufficient to obtain either the normal p_N or the tangential component p_T , which are tensor elements tangential or normal to surfaces of same densities. E.g., p_T can be found from p_N through the

condition of mechanical equilibrium [45]

$$p_T = p_N + \frac{r}{2} \cdot \frac{dp_N}{dr}. \quad (21)$$

Here, the normal pressure component was calculated by simulation. Exemplarily, simulation results are shown in figure 13 at $T^* = 0.8$ for three specific droplet sizes.

In analogy to the density profile an expression following [46, 53]

$$p_N = \frac{1}{2}(p_l + p_v) - \frac{1}{2}(p_l - p_0) \tanh\left(\frac{2}{D_l}(r - r_l)\right) + \frac{1}{2}(p_v - p_0) \tanh\left(\frac{2}{D_v}(r - r_v)\right), \quad (22)$$

was fitted to simulation data, wherein p_l , p_v , and p_0 characterize the normal pressure in the bulk liquid, the bulk vapour and the pressure minimum. Furthermore, r_l and r_v denote the radii where the normal pressure profile has inflection points on the liquid and vapour side of the fluid. The width of the normal pressure profile on the liquid and the vapour side of the interface is denoted by D_l and D_v , respectively.

As for the to the density profiles, equation (22) was also fitted first to all 88 simulation data sets of normal pressure individually. All parameters can be found in table 6.

Subsequently, a generalized normal pressure profile was determined, being both temperature and size dependent, i.e. $p_N(T, R_e)$.

The pressures in the liquid and vapour far away from the interface were correlated by

$$p_l^* = p_\sigma^* + \frac{0.3323 + 6.0260 \cdot (T_c^* - T^*)^2}{R_e^* - 9.1020 + 1.2910 \cdot (T_c^* - T^*)^{-1} + 15.3600 \cdot (T_c^* - T^*)}, \quad (23)$$

and

$$p_v^* = p_\sigma^* + \frac{-0.0272 + 0.0829 \cdot T^{*2}}{R_e^* - 2.9110 + 0.5257 \cdot (T_c^* - T^*)^{-1} + 2.9270 \cdot (T_c^* - T^*)}, \quad (24)$$

where p_σ^* is taken from homogeneous VLE results, cf. equation (3). For $R_e^* \rightarrow \infty$, both p_l^* and p_v^* are identical with p_σ^* . The properties r_l^* , r_v^* , D_l^* , and D_v^* are linear functions of the radius of the droplet R_e^* , where the same form was chosen as for the generalized density correlation, cf. equation (20). All parameters are given in table 7. Here, p_0^* is depends exponentially on the temperature and was correlated using

$$p_0^* = -0.4713 + 0.2101 \cdot T^* \quad (25)$$

$$+ (0.3265 - 0.0441 \cdot T^*) \cdot (1 - \exp(-(0.4240 - 0.3350 \cdot T^*) \cdot R_e^*)).$$

Figure 14 shows the normal pressure profile results from generalized correlation for $R_e^* = 7, 11$, and 16 at two different temperatures.

To calculate the surface tension of spherical interfaces from the normal pressure the following expression was applied [45]

$$\gamma^3 = -\frac{1}{8}(p_l - p_v)^2 \int_0^\infty dr r^3 \frac{dp_N}{dr}. \quad (26)$$

With equations (20) and (22) to (25), the surface tension γ^* can be calculated using equation (26). Figure 15 shows these results. The surface tension is presented as a function of the droplet size and temperature. It can be seen that the surface tension is rising in monotonous way, with the surface tension of the planar surface $R_e^* \rightarrow \infty$ as the supreme limiting value. But should be noted that the surface tension is reduced between 15% ($T^* = 0.65$) and 60% ($T^* = 0.95$) for the smallest simulated droplets.

The reported result of Thompson et al. [45] at $T^* = 0.8$ is 14% higher for the thermodynamic route and 8% for the mechanical route. The result of Bardouni et al. [46] is questionable as it is above the planar interface value.

The presented generalized correlations are valid for droplets with $R_e^* < 20$. For larger droplets, the generalized normal pressure representation fails in equation (26), since the surface tension is almost only dependent on the very small pressure difference between the liquid and the vapour pressure. This pressure difference tends to zero for big droplets, where the extrapolation of the fits breaks down. But a continuation to large radii should be possible, since the surface tension of the planar surface is known.

Laplace equation [2]

$$R_\gamma = \frac{2\gamma}{p_l - p_v}, \quad (27)$$

degenerates in the present context to a definition of a radius R_γ , i.e. surface of tension. Values for R_γ^* from individual droplet simulation runs are given in table 6 and are shown in figures 11 and 13. But the surface of tension can also be obtained from the generalized normal pressure correlation, i.e. equations (20) and (22) to (25), using equation (27). Results from this route, agreeing well to individual simulation data, are depicted in figures 13 and 14.

Finally, the Tolman length is defined as the difference of Gibbs absorption dividing surface and surface of tension

$$\delta = R_e - R_\gamma. \quad (28)$$

It was obtained from individual normal pressure profiles, given in table 6. The Tolman length ranges from $\delta^* = 1$ for cold droplets ($T^* = 0.65$), where it is independent on droplet size, to $\delta^* = 5$ for the smallest droplet at the highest temperature ($T^* = 0.95$), where it is size dependent. A comparison of the results from individual and generalized normal pressure profiles in figure 16 shows, despite some deviations in the order of 0.2 in absolute terms, the consistency of the presented data and generalized correlations.

7 Conclusion

It can be concluded that the Lennard-Jones potential, truncated and shifted at 2.5σ , is a good molecular model for Argon, Krypton, and Xenon as well as for Methane to describe VLE and to predict surface properties, i.e. the surface tension. This potential was investigated comprehensively, covering a wide range of states, and an extensive data base is presented. For planar interfaces, it was found that the classical profile is well suited to describe the density variation through the interface. A thorough comparison to data of other authors was made, where generally good agreement is found. Furthermore, a consistent set of correlations representing this data, to be used as tools in subsequent work, was developed. For convenient use, a FORTRAN 77 code is available in the internet [54].

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Table 1. Vapour-liquid equilibria from homogeneous bulk VLE simulations of Lennard-Jones fluid truncated and shifted at 2.5σ . The number in parentheses denotes the uncertainty in the last digit.

T^*	p_σ^*	ρ^*	ρ''^*	Δh_v^*
0.64	0.00217 (4)	0.8176 (1)	0.00351 (4)	5.7118 (9)
0.67	0.00335 (4)	0.8024 (1)	0.00525 (4)	5.5910 (8)
0.70	0.00479 (4)	0.7866 (1)	0.00727 (4)	5.4666 (9)
0.73	0.00697 (4)	0.7704 (1)	0.01036 (8)	5.325 (1)
0.76	0.00944 (8)	0.7538 (2)	0.01374 (9)	5.179 (1)
0.79	0.01241 (8)	0.7361 (2)	0.01776 (9)	5.022 (2)
0.82	0.01640 (8)	0.7181 (2)	0.0233 (1)	4.844 (2)
0.85	0.0214 (2)	0.6986 (2)	0.0303 (3)	4.639 (3)
0.88	0.0274 (2)	0.6784 (2)	0.0392 (4)	4.413 (5)
0.91	0.0336 (3)	0.6556 (3)	0.0483 (8)	4.172 (8)
0.94	0.0417 (3)	0.6309 (4)	0.0616 (8)	3.87 (1)
0.97	0.0504 (4)	0.6032 (5)	0.0763 (9)	3.56 (2)
1.00	0.0606 (8)	0.5712 (7)	0.096 (2)	3.18 (3)
1.03	0.073 (1)	0.530 (1)	0.127 (4)	2.63 (5)
1.06	0.0855 (8)	0.463 (3)	0.168 (4)	1.88 (8)

Table 2. Vapour and liquid densities as well as interface properties from planar interface VLE simulations. The number in parentheses denotes the uncertainty in the last digit.

T^*	l^*	N	ρ_l^*	ρ_v^*	D^*	γ^*
0.625	11.85	2759	0.8244	0.0024	1.5658	0.727 (9)
0.650	11.91	2769	0.8120	0.0036	1.6616	0.675 (9)
0.675	11.97	2769	0.7994	0.0051	1.7494	0.633 (9)
0.700	12.04	2776	0.7863	0.0072	1.8621	0.581 (8)
0.725	12.11	2782	0.7729	0.0095	1.9844	0.549 (9)
0.750	12.18	2797	0.7599	0.0122	2.1065	0.493 (8)
0.775	12.26	2852	0.7452	0.0153	2.2517	0.455 (8)
0.800	12.34	2872	0.7307	0.0203	2.4348	0.403 (7)
0.825	12.43	2888	0.7156	0.0244	2.5695	0.369 (7)
0.850	12.52	2897	0.6993	0.0302	2.7943	0.322 (7)
0.875	12.63	2961	0.6821	0.0368	3.0027	0.275 (6)
0.900	12.75	2958	0.6632	0.0443	3.3181	0.239 (6)
0.925	12.87	3065	0.6437	0.0508	3.6935	0.191 (6)
0.950	13.01	3141	0.6222	0.0635	4.0550	0.156 (6)
0.975	13.18	3315	0.5987	0.0793	4.6938	0.122 (5)
1.000	13.39	3538	0.5672	0.0891	5.4502	0.081 (5)
1.025	13.66	3597	0.5265	0.1064	7.1065	0.057 (4)
1.050	14.05	4027	0.4827	0.1233	8.1196	0.030 (4)

Table 3. Lennard-Jones 2.5 truncated and shifted potential parameters for real substances adjusted to vapour pressures and bubble densities from equations of state [40–42].

Fluid	σ Å	ϵ/k_B K	Equation of state
Ar	3.3916	137.90	[40]
Kr	3.6233	191.52	[41]
Xe	3.9450	265.78	[41]
CH ₄	3.7241	175.06	[42]

Table 4. Vapor and liquid densities as well as interface properties from spherical interface VLE simulations and individual correlations, cf. equation (16).

N	N_L	ρ_l^*	ρ_v^*	ρ_0^*	$D_{\rho l}^*$	$r_{\rho l}^*$	$D_{\rho v}^*$	$r_{\rho v}^*$	R_e^*
$T^* = 0.65$									
2255	1083	0.8282	0.00611	−0.127	1.988	6.869	2.230	7.587	6.85
2851	1725	0.8264	0.00585	−0.124	2.043	8.022	2.356	8.777	7.99
3646	2556	0.8250	0.00562	−0.131	2.057	9.149	2.377	9.891	9.10
4443	3378	0.8239	0.00548	−0.123	2.067	10.032	2.446	10.778	9.99
5385	4365	0.8231	0.00524	−0.121	2.087	10.926	2.458	11.691	10.88
6429	5409	0.8228	0.00522	−0.129	2.120	11.740	2.494	12.513	11.67
7524	6500	0.8222	0.00524	−0.120	2.117	12.467	2.578	13.214	12.41
8819	7803	0.8220	0.00514	−0.125	2.129	13.257	2.543	14.024	13.19
9315	8307	0.8215	0.00513	−0.130	2.153	13.531	2.619	14.254	13.47
9513	8450	0.8213	0.00515	−0.127	2.140	13.633	2.624	14.351	13.57
10109	9105	0.8211	0.00509	−0.119	2.129	13.940	2.656	14.645	13.89
10806	9810	0.8211	0.00506	−0.128	2.174	14.294	2.724	14.990	14.24
11499	10499	0.8207	0.00509	−0.129	2.153	14.624	2.686	15.330	14.56
$T^* = 0.70$									
2759	410	0.8065	0.01240	−0.100	2.154	4.976	2.592	5.662	5.04
3058	1018	0.8030	0.01071	−0.109	2.210	6.778	2.664	7.488	6.79
3357	1367	0.8018	0.01047	−0.113	2.224	7.481	2.706	8.168	7.49
4151	2282	0.7999	0.00980	−0.117	2.273	8.886	2.774	9.591	8.87
4897	3036	0.7991	0.00977	−0.115	2.302	9.769	2.869	10.457	9.71
5841	4041	0.7980	0.00943	−0.120	2.312	10.747	2.913	11.408	10.72
6977	5224	0.7971	0.00920	−0.117	2.344	11.699	3.066	12.325	11.62
8015	6279	0.7969	0.00907	−0.148	2.399	12.441	3.095	12.972	12.41
9099	7378	0.7961	0.00905	−0.121	2.359	13.129	3.087	13.752	13.10
10286	8560	0.7957	0.00907	−0.118	2.370	13.794	3.110	14.425	13.76
11081	9065	0.7957	0.00909	−0.128	2.400	14.065	3.148	14.688	14.02
11668	9980	0.7953	0.00890	−0.112	2.387	14.510	3.224	15.118	14.48
12557	10864	0.7951	0.00898	−0.124	2.408	14.933	3.225	15.541	14.89
13351	11682	0.7950	0.00885	−0.123	2.399	15.303	3.164	15.935	15.26
14243	12569	0.7948	0.00890	−0.120	2.420	15.672	3.288	16.270	15.64
15035	13372	0.7947	0.00887	−0.118	2.454	15.992	3.375	16.576	15.96
15726	14067	0.7945	0.00887	−0.130	2.418	16.274	3.270	16.846	16.23
17219	15574	0.7941	0.00883	−0.125	2.428	16.827	3.345	17.369	16.79

Table 4: continued.

$T^* = 0.75$									
4700	1479	0.7753	0.01705	−0.084	2.531	7.708	3.385	8.330	7.78
5362	2275	0.7734	0.01636	−0.101	2.563	8.938	3.402	9.523	8.97
6079	3079	0.7722	0.01591	−0.121	2.594	9.901	3.434	10.431	9.92
6926	3982	0.7708	0.01565	−0.150	2.650	10.785	3.586	11.179	10.80
7799	4906	0.7708	0.01543	−0.110	2.648	11.560	3.631	12.073	11.58
8632	5788	0.7697	0.01520	−0.118	2.653	12.229	3.621	12.747	12.23
9416	6611	0.7694	0.01502	−0.137	2.684	12.782	3.661	13.224	12.78
10296	7532	0.7690	0.01485	−0.129	2.703	13.338	3.795	13.378	13.35
11227	8475	0.7687	0.01479	−0.129	2.698	13.885	3.731	14.343	13.88
12009	9282	0.7684	0.01472	−0.113	2.687	14.314	3.758	14.829	14.31
12788	10080	0.7683	0.01468	−0.109	2.690	14.705	3.870	15.183	14.72
$T^* = 0.80$									
5945	849	0.7513	0.02707	−0.102	2.931	6.441	3.818	7.040	6.56
6335	1577	0.7465	0.02532	−0.097	2.874	7.979	3.691	8.574	8.06
6726	2024	0.7449	0.02508	−0.095	2.936	8.655	4.025	9.091	8.75
7401	2808	0.7441	0.02457	−0.107	2.979	9.669	4.118	10.069	9.75
8370	3878	0.7422	0.02412	−0.133	3.016	10.778	4.177	11.054	10.85
9429	5057	0.7418	0.02351	−0.119	3.036	11.816	4.098	12.288	11.85
10704	6388	0.7408	0.02347	−0.108	3.007	12.780	4.076	13.272	12.81
11766	7509	0.7402	0.02321	−0.121	3.077	13.477	4.315	13.852	13.52
12546	8332	0.7399	0.02311	−0.125	3.063	13.956	4.279	14.314	13.99
13124	8974	0.7395	0.02281	−0.129	3.121	14.304	4.407	14.638	14.34
13801	9665	0.7397	0.02281	−0.106	3.073	14.677	4.328	15.154	14.70
$T^* = 0.85$									
5992	1369	0.7171	0.03847	−0.073	3.369	7.612	4.505	8.079	7.79
6616	2170	0.7141	0.03717	−0.101	3.444	8.926	4.670	9.250	8.98
7225	2963	0.7123	0.03581	−0.115	3.447	9.944	4.656	10.246	10.07
8012	3792	0.7120	0.03559	−0.095	3.443	10.831	4.511	11.295	10.87
8721	4564	0.7102	0.03543	−0.153	3.586	11.507	4.869	11.712	11.62
9675	5583	0.7096	0.03529	−0.145	3.572	12.335	4.894	12.599	12.42
10526	6477	0.7098	0.03511	−0.111	3.506	12.977	4.757	13.346	13.05
11369	7381	0.7083	0.03509	−0.159	3.691	13.542	5.098	13.752	13.63
12035	8087	0.7094	0.03453	−0.109	3.564	13.993	4.407	14.431	14.05
12688	8804	0.7082	0.03467	−0.121	3.644	14.370	5.282	14.618	14.45
13352	9518	0.7098	0.03423	−0.133	3.640	14.790	4.825	15.239	14.82

Table 4: continued.

$T^* = 0.90$									
5913	1340	0.6830	0.05580	−0.089	4.041	7.575	5.370	7.901	7.84
6460	2064	0.6800	0.05414	−0.097	3.981	8.863	4.998	9.321	9.05
7112	2850	0.6791	0.05309	−0.106	4.123	9.900	5.235	10.305	10.08
7848	3693	0.6778	0.05241	−0.114	4.206	10.806	5.464	11.106	11.00
8591	4541	0.6763	0.05194	−0.123	4.172	11.629	5.412	11.963	11.78
9334	5348	0.6765	0.05198	−0.133	4.247	12.312	5.499	12.706	12.43
10074	6218	0.6749	0.05118	−0.167	4.343	12.915	5.833	13.091	13.08
10805	7015	0.6735	0.05093	−0.138	4.298	13.449	5.894	13.573	13.62
11727	8005	0.6749	0.05142	−0.174	4.399	14.109	5.864	14.398	14.22
12373	8752	0.6752	0.05095	−0.195	4.401	14.499	5.982	14.643	14.64
13100	9448	0.6736	0.05163	−0.168	4.442	14.905	5.927	15.137	15.04
14116	10553	0.6727	0.05196	−0.173	4.377	15.489	5.847	15.720	15.60
14760	11287	0.6708	0.05279	−0.251	4.661	15.814	6.385	15.921	15.96
$T^* = 0.95$									
7799	1902	0.6408	0.07841	−0.051	5.008	8.497	5.991	8.857	8.93
8409	2727	0.6404	0.07666	−0.074	5.073	9.726	6.066	10.132	10.08
8844	3353	0.6384	0.07457	−0.069	5.059	10.498	6.017	10.902	10.72
9280	3853	0.6383	0.07430	−0.043	4.984	11.011	5.838	11.407	11.56
9890	4546	0.6374	0.07408	−0.055	5.069	11.678	6.048	12.060	12.02
10232	4849	0.6384	0.07516	−0.059	5.112	11.940	5.989	12.363	12.23
10661	5440	0.6360	0.07469	−0.151	5.425	12.443	6.774	12.714	12.72
11190	6093	0.6370	0.07313	−0.093	5.205	12.952	6.378	13.284	13.06
11535	6524	0.6361	0.07280	−0.108	5.295	13.284	6.551	13.646	13.21
12150	7246	0.6360	0.07157	−0.045	5.049	13.734	6.082	13.987	13.97
12878	8050	0.6352	0.07039	−0.097	4.895	14.185	4.952	14.626	14.50

Table 5. Parameters for the spherical interface density profile, cf. equation (20).

Property	a_1	a_2	a_3	a_4	a_5	a_6
ρ_0^*	−0.0916	−0.1743	0.0208	0.0232	−0.0355	0
$r_{\rho l}^*$	0.0004	0.3612	−0.0702	1.0030	−0.0092	0.0035
$r_{\rho v}^*$	0.4465	1.2110	−0.0734	1.1010	−0.2001	0.0126
$D_{\rho l}^*$	1.7520	−1.7910	0.3830	−0.0701	0.1321	0
$D_{\rho v}^*$	5.9180	−10.0800	0.0641	0.0149	0.0718	0

Table 6. Vapor and liquid pressures as well as interface properties from spherical interface VLE simulations and individual correlations, cf. equation (22).

N	p_l^*	p_v^*	p_0^*	D_l^*	r_l^*	D_v^*	r_v^*	R_γ^*	δ^*	γ^*
$T^* = 0.65$										
2255	0.2033	0.00374	−0.136	2.349	6.107	2.207	6.794	5.74	1.11	0.572
2851	0.1727	0.00359	−0.086	2.229	7.165	2.363	7.801	6.91	1.08	0.585
3646	0.1504	0.00344	−0.076	2.238	8.351	2.264	9.023	8.06	1.04	0.592
4443	0.1398	0.00338	−0.064	2.259	9.175	2.523	9.794	8.94	1.05	0.610
5385	0.1289	0.00324	−0.091	2.402	10.209	2.387	10.810	9.83	1.05	0.618
6429	0.1187	0.00322	−0.073	2.346	10.985	2.418	11.576	10.66	1.01	0.615
7524	0.1122	0.00324	−0.061	2.310	11.681	2.496	12.255	11.40	1.01	0.621
8819	0.1064	0.00316	−0.068	2.376	12.517	2.445	13.095	12.18	1.01	0.629
9315	0.1038	0.00317	−0.056	2.351	12.735	2.567	13.306	12.46	1.01	0.627
9513	0.1026	0.00317	−0.054	2.315	12.845	2.522	13.415	12.57	1.00	0.625
10109	0.1005	0.00315	−0.055	2.316	13.178	2.468	13.749	12.89	1.00	0.628
10806	0.0973	0.00310	−0.044	2.294	13.469	2.613	14.026	13.25	0.99	0.624
11499	0.0960	0.00314	−0.044	2.315	13.801	2.644	14.374	13.56	1.00	0.630
$T^* = 0.70$										
2759	0.2327	0.00781	−0.154	2.440	4.096	2.364	4.825	3.71	1.33	0.417
3058	0.1779	0.00685	−0.104	2.383	5.864	2.495	6.518	5.55	1.24	0.474
3357	0.1627	0.00669	−0.107	2.451	6.622	2.509	7.260	6.26	1.23	0.488
4151	0.1383	0.00630	−0.093	2.511	8.065	2.528	8.705	7.68	1.19	0.507
4897	0.1272	0.00626	−0.101	2.619	9.016	2.592	9.629	8.56	1.15	0.518
5841	0.1141	0.00609	−0.076	2.512	9.961	2.586	10.567	9.57	1.15	0.517
6977	0.1052	0.00594	−0.068	2.541	10.919	2.626	11.527	10.54	1.08	0.523
8015	0.0994	0.00586	−0.065	2.574	11.638	2.727	12.205	11.28	1.13	0.527
9099	0.0954	0.00582	−0.062	2.574	12.315	2.768	12.873	11.96	1.14	0.535
10286	0.0902	0.00586	−0.052	2.510	12.948	2.803	13.481	12.64	1.12	0.533
11081	0.0882	0.00585	−0.053	2.557	13.233	2.826	13.759	12.91	1.11	0.532
11668	0.0864	0.00575	−0.047	2.553	13.652	2.883	14.189	13.36	1.12	0.539
12557	0.0836	0.00578	−0.044	2.560	14.072	2.939	14.611	13.78	1.11	0.536
13351	0.0817	0.00569	−0.040	2.534	14.446	2.872	15.027	14.15	1.11	0.538
14243	0.0794	0.00571	−0.048	2.550	14.876	2.773	15.409	14.54	1.10	0.536
15035	0.0780	0.00570	−0.028	2.561	15.141	2.877	15.862	14.86	1.10	0.537
15726	0.0770	0.00568	−0.050	2.590	15.521	2.718	16.087	15.14	1.09	0.540
17219	0.0738	0.00569	−0.033	2.503	15.949	2.996	16.451	15.71	1.08	0.535

Table 6: continued.

$T^* = 0.75$										
4700	0.1358	0.01121	−0.079	2.595	6.709	2.954	7.297	6.29	1.49	0.397
5362	0.1197	0.01084	−0.069	2.727	7.939	3.151	8.539	7.62	1.35	0.413
6079	0.1097	0.01054	−0.051	2.779	8.998	2.795	9.850	8.60	1.32	0.425
6926	0.1000	0.01041	−0.056	2.727	9.824	3.134	10.377	9.50	1.30	0.425
7799	0.0951	0.01026	−0.069	2.837	10.694	3.022	11.252	10.28	1.30	0.435
8632	0.0897	0.01010	−0.044	2.744	11.223	3.290	11.747	10.94	1.29	0.435
9416	0.0860	0.01002	−0.055	2.767	11.893	3.011	12.438	11.49	1.29	0.437
10296	0.0825	0.00990	−0.048	2.731	12.407	3.151	12.893	12.06	1.29	0.438
11227	0.0798	0.00989	−0.043	2.797	12.939	3.243	13.467	12.59	1.29	0.441
12009	0.0782	0.00982	−0.052	2.872	13.479	3.057	14.060	13.03	1.28	0.445
12788	0.0758	0.00979	−0.043	2.743	13.760	3.217	14.213	13.45	1.27	0.444
$T^* = 0.80$										
5945	0.1408	0.01792	−0.136	3.103	5.427	3.185	6.040	4.79	1.77	0.294
6335	0.1194	0.01697	−0.127	3.218	7.024	3.270	7.596	6.34	1.72	0.325
6726	0.1072	0.01686	−0.076	2.941	7.556	3.411	8.053	7.12	1.63	0.322
7401	0.0989	0.01656	−0.076	3.078	8.649	3.368	9.179	8.14	1.61	0.335
8370	0.0880	0.01631	−0.052	2.985	9.690	3.542	10.159	9.31	1.54	0.334
9429	0.0822	0.01601	−0.052	3.008	10.746	3.455	11.209	10.34	1.51	0.342
10704	0.0778	0.01592	−0.036	3.025	11.719	3.514	12.310	11.29	1.52	0.349
11766	0.0735	0.01578	−0.023	2.936	12.319	3.834	12.799	12.05	1.47	0.348
12546	0.0728	0.01572	−0.033	3.133	12.916	3.636	13.514	12.47	1.52	0.356
13124	0.0702	0.01553	−0.038	3.103	13.273	3.558	13.760	12.86	1.48	0.352
13801	0.0693	0.01556	−0.037	3.107	13.650	3.584	14.150	13.22	1.48	0.355
$T^* = 0.85$										
5992	0.1072	0.02560	−0.093	3.522	6.344	3.916	6.886	5.60	2.19	0.229
6616	0.0930	0.02497	−0.076	3.391	7.689	3.865	8.148	7.05	1.93	0.240
7225	0.0857	0.02429	−0.071	3.440	8.766	3.861	9.215	8.12	1.95	0.250
8012	0.0826	0.02424	−0.064	3.597	9.567	4.091	10.015	8.95	1.92	0.261
8721	0.0777	0.02406	−0.045	3.441	10.247	4.023	10.692	9.73	1.89	0.261
9675	0.0736	0.02399	−0.033	3.350	11.118	4.149	11.645	10.56	1.86	0.262
10526	0.0705	0.02392	−0.037	3.360	11.749	4.042	12.149	11.25	1.80	0.262
11369	0.0692	0.02380	−0.055	3.707	12.471	4.070	12.919	11.76	1.87	0.267
12035	0.0668	0.02373	−0.038	3.574	12.755	4.220	13.143	12.23	1.82	0.263
12688	0.0660	0.02356	−0.033	3.554	13.164	4.169	13.587	12.65	1.80	0.268
13352	0.0664	0.02344	−0.049	3.788	13.689	4.191	14.128	13.01	1.81	0.280

Table 6: continued.

$T^* = 0.90$										
5913	0.0917	0.03638	−0.101	3.954	6.146	4.390	6.578	4.86	2.98	0.134
6460	0.0886	0.03570	−0.123	4.318	7.603	4.512	8.041	6.25	2.80	0.165
7112	0.0759	0.03523	−0.042	3.826	8.408	4.609	8.795	7.56	2.52	0.154
7848	0.0763	0.03497	−0.061	4.240	9.377	4.751	9.792	8.39	2.61	0.173
8591	0.0719	0.03465	−0.042	4.039	10.159	4.705	10.544	9.33	2.45	0.174
9334	0.0712	0.03469	−0.027	4.238	10.822	4.986	11.338	9.92	2.51	0.181
10074	0.0694	0.03421	−0.043	4.247	11.558	4.824	11.970	10.68	2.40	0.188
10805	0.0686	0.03417	−0.039	4.527	12.110	4.989	12.623	11.02	2.60	0.191
11727	0.0671	0.03425	−0.058	4.666	12.832	5.101	13.229	11.74	2.48	0.193
12373	0.0661	0.03380	−0.018	4.201	13.137	5.012	13.592	12.45	2.19	0.202
13100	0.0640	0.03432	−0.039	4.373	13.510	4.969	13.867	12.65	2.39	0.188
14116	0.0635	0.03431	−0.046	4.464	14.162	4.981	14.540	13.17	2.43	0.193
14760	0.0620	0.03426	−0.055	4.811	14.627	5.291	15.013	13.47	2.49	0.189
$T^* = 0.95$										
7799	0.0795	0.04916	−0.092	4.849	6.588	5.469	6.872	3.84	5.09	0.059
8409	0.0772	0.04860	−0.024	5.162	7.782	6.147	8.289	5.71	4.37	0.082
8844	0.0757	0.04817	−0.080	5.328	8.626	5.897	8.917	6.68	4.04	0.092
9280	0.0727	0.04805	−0.055	4.916	9.130	5.522	9.398	7.62	3.94	0.094
9890	0.0712	0.04800	−0.052	5.173	9.998	5.750	10.330	8.12	3.89	0.095
10232	0.0721	0.04837	−0.050	5.113	10.083	5.734	10.393	8.41	3.82	0.100
10661	0.0706	0.04796	−0.053	5.357	10.541	5.940	10.835	8.93	3.79	0.102
11190	0.0701	0.04780	−0.049	5.373	11.093	6.001	11.390	9.54	3.52	0.107
11535	0.0693	0.04759	−0.051	5.469	11.581	5.989	11.928	9.77	3.44	0.107
12150	0.0692	0.04750	−0.044	5.498	12.175	6.106	12.541	10.43	3.54	0.114
12878	0.0666	0.04751	−0.031	5.100	12.525	5.822	12.826	11.08	3.42	0.106

Table 7. Parameters for the spherical interface normal pressure profile, cf. equation (20).

Property	a_1	a_2	a_3	a_4	a_5	a_6
r_l^*	−0.7686	1.6740	−0.3063	0.8764	0.1710	0.0048
r_v^*	−0.1599	2.3170	−0.3452	0.8438	0.1965	0.0053
D_l^*	1.2740	0.1677	0.3957	−0.1126	0.1621	0.0038
D_v^*	2.4040	−2.8670	0.3418	−0.0393	0.1007	0.0046

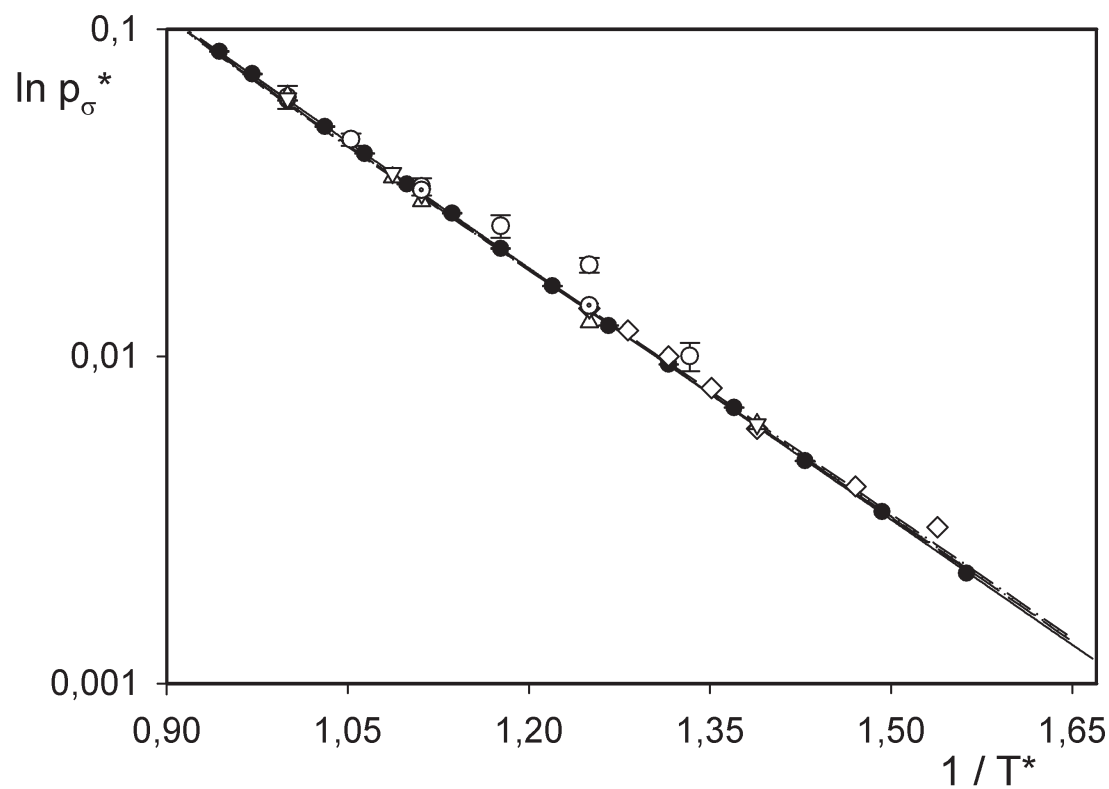


Figure 1. Logarithmic vapour pressure vs. inverse temperature for the truncated and shifted Lennard-Jones fluid: ● results from homogeneous bulk VLE simulations (this work), – correlation, cf. equation (3), simulation results of other authors: ○ Camp and Allen [9], ◇ Mareschal et al. [5], △ Trokhymchuk and Alejandre (Monte-Carlo) [12], ▽ Trokhymchuk and Alejandre (molecular dynamics) [12], ⊙ Adams and Henderson [6]. Experimental real substance data reduced with the potential parameters from table 3: – · – Argon [40], · · · Krypton [41], - - - Xenon [41], – – Methane [42] (hardly discernable from correlation).

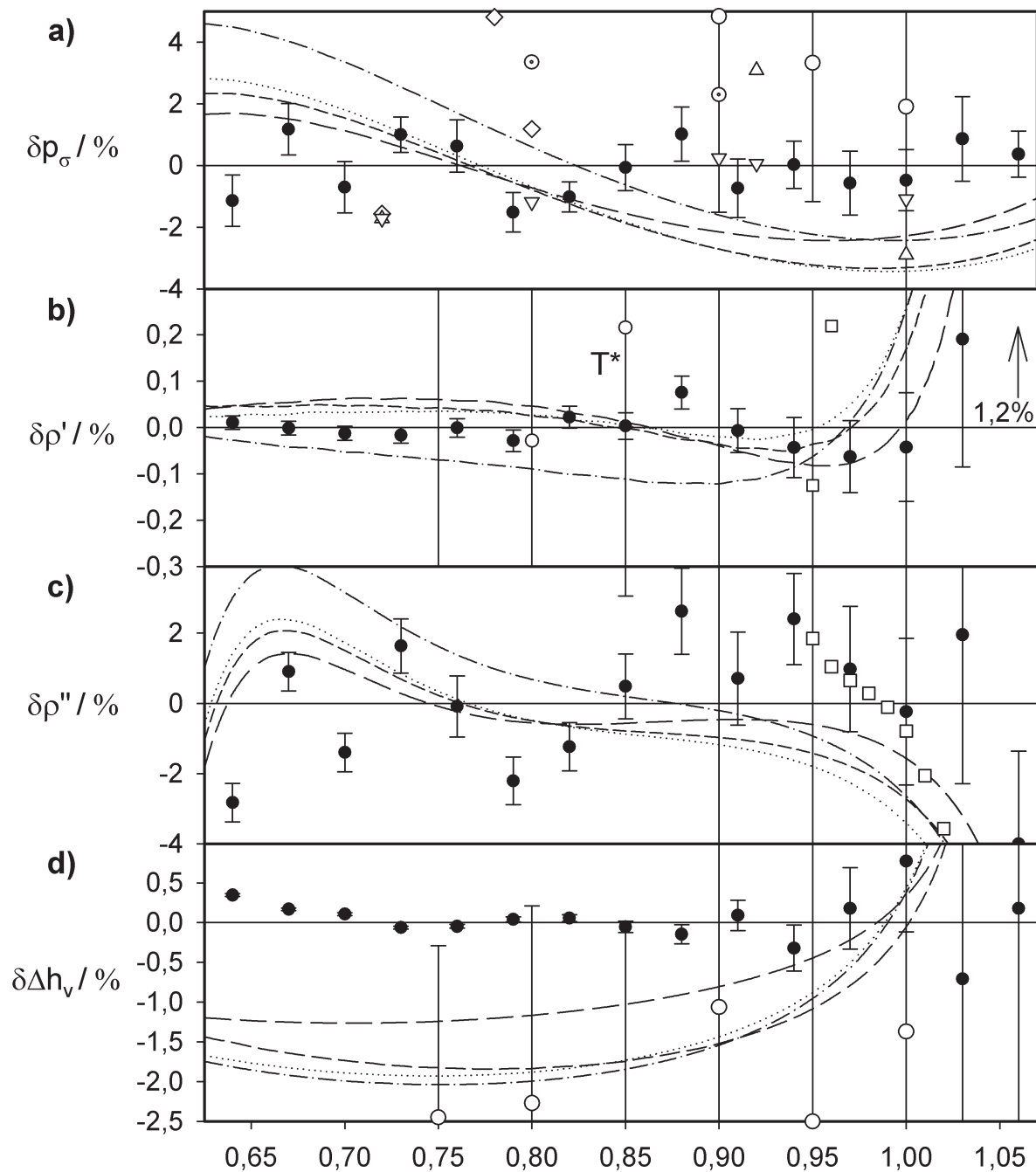


Figure 2. Deviation plots for homogeneous VLE properties. The Baselines are the present correlations, cf. equations (3) to (6): ● this work, □ Shi and Johnson [7], ○ Camp and Allen [9], ◇ Mareschal et al. [5], △ Trokhymchuk and Alejandre (Monte-Carlo) [12], ▽ Trokhymchuk and Alejandre (molecular dynamics) [12], ○ Adams and Henderson [6]. Experimental real substance data reduced with the potential parameters from table 3: - · - Argon [40], · · · Krypton [41], - - - Xenon [41], - - Methane [42].

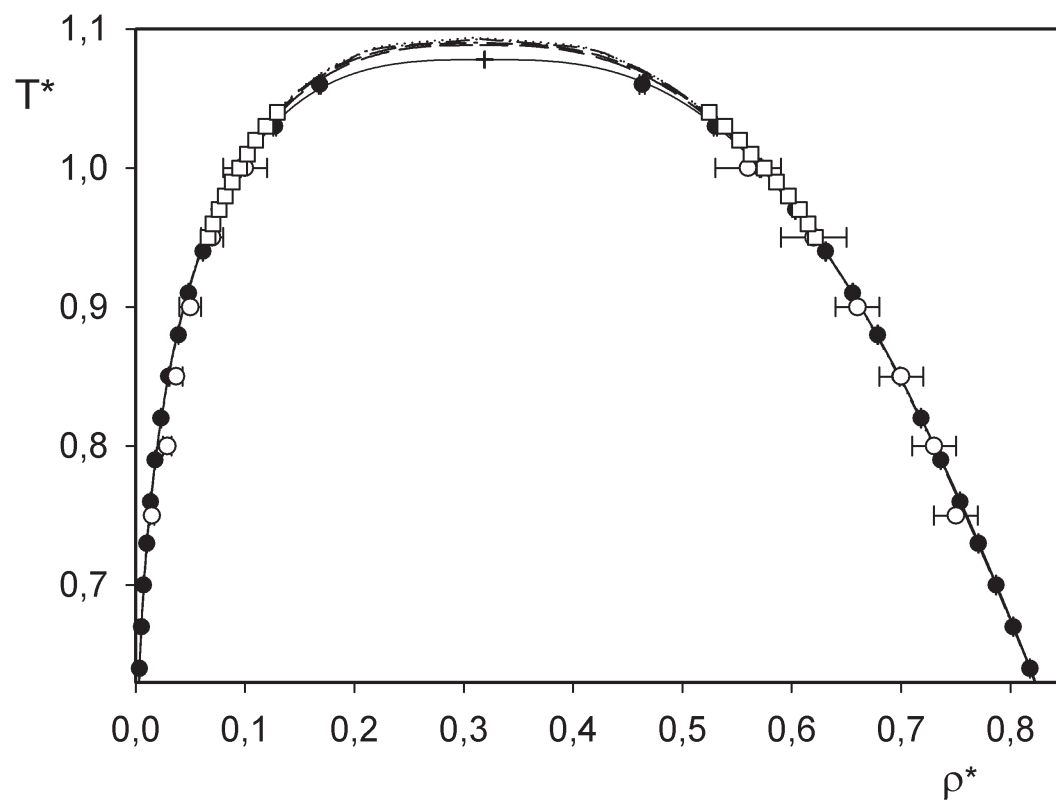


Figure 3. Temperature vs. saturated densities for the truncated and shifted Lennard-Jones fluid: ● results from homogeneous bulk VLE simulations (this work), – correlations, cf. equations (4) and (5), + critical point, simulation results of other authors: □ Shi and Johnson [7], ○ Camp and Allen [9]. Experimental real substance data reduced with the potential parameters from table 3: – · – Argon [40], · · · Krypton [41], - - - Xenon [41], - - - Methane [42] (deviations from correlation only discernable in the critical region).

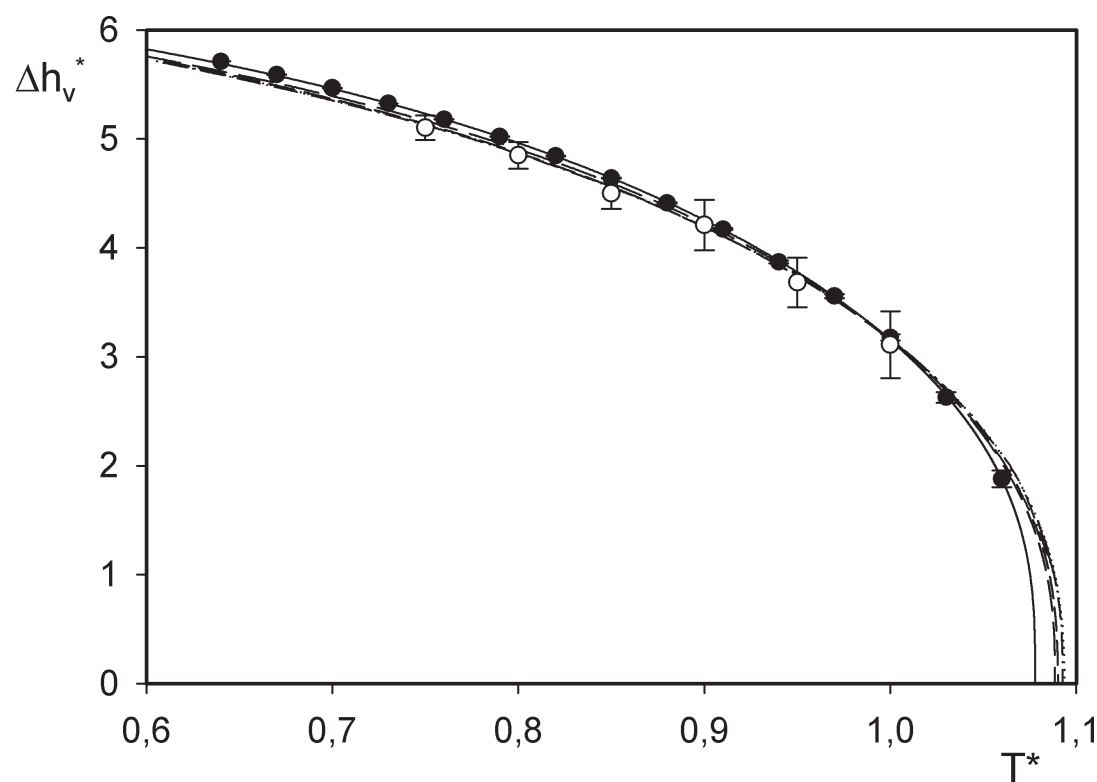


Figure 4. Enthalpy of vaporization vs. temperature for the truncated and shifted Lennard-Jones fluid: ● results from homogeneous bulk VLE simulations (this work), — correlation, cf. equation (6), simulation results of other authors: ○ Camp and Allen [9]. Experimental real substance data reduced with the potential parameters from table 3: - · - Argon [40], · · · Krypton [41], - - - Xenon [41], - - Methane [42].

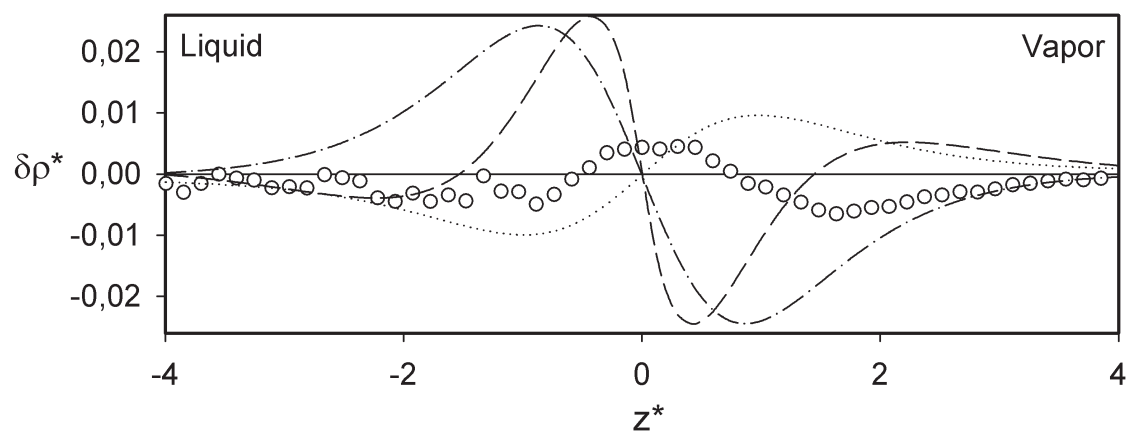


Figure 5. Deviation plot for the density profile near the interface of the truncated and shifted Lennard-Jones fluid at $T^* = 0.8$. Different correlations are compared to results from planar interface VLE simulations (this work): $\circ$. The baseline is the Classical profile [25], $- \cdot -$ Fisk-Widom profile [26], $\cdots$ Error-function profile [27], $- - -$ Exponential profile [28].

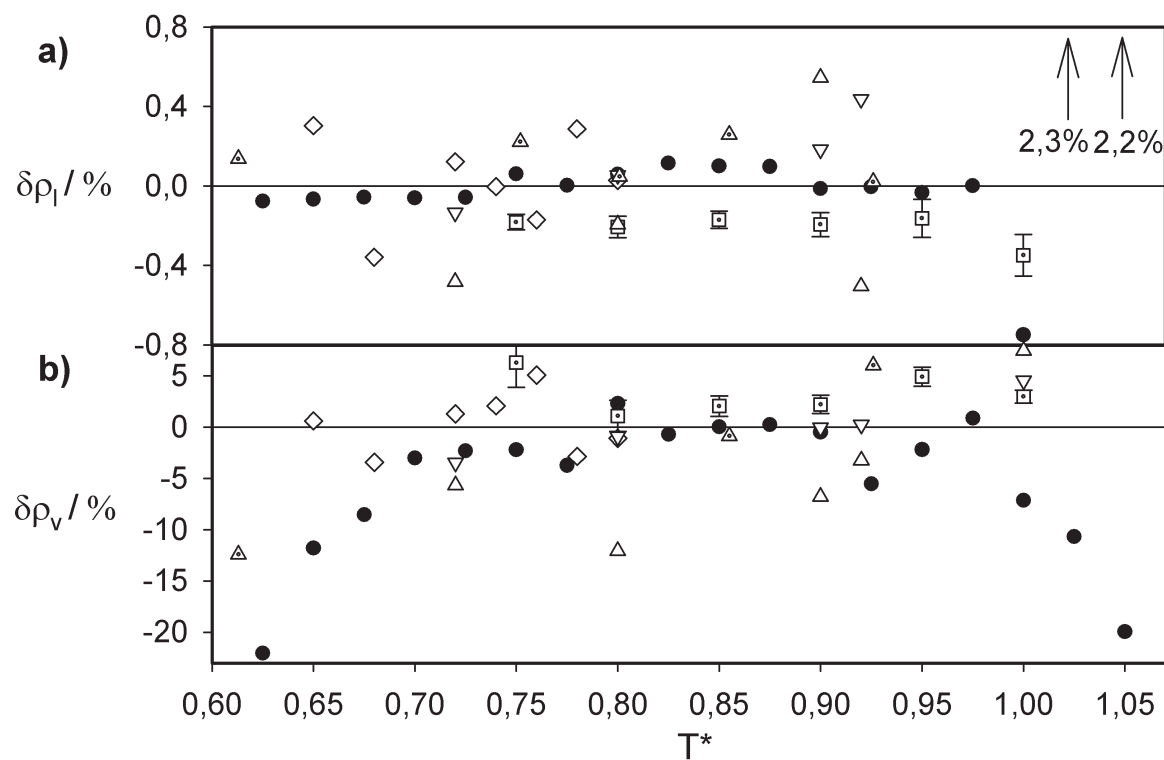


Figure 6. Deviation plots for saturated densities from planar interface VLE simulations. The baselines are the correlations for saturated densities from homogeneous VLE calculations, cf. equations (4) and (5): ● this work, ◇ Mareschal et al. [5], □ Haye et al. [10], △ Trokhymchuk and Alejandre (Monte-Carlo) [12], ▽ Trokhymchuk and Alejandre (molecular dynamics) [12], ‘dotted triangle up’ Dunikov et al. [11].

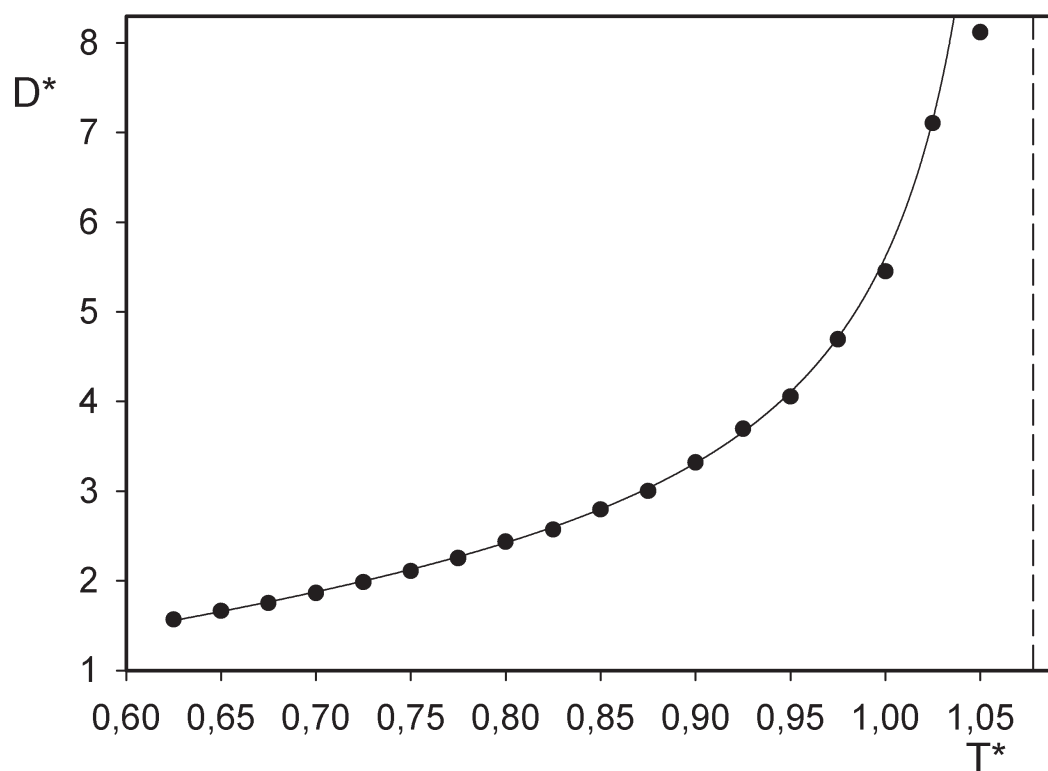


Figure 7. Interface width vs. temperature from planar interface VLE simulations: ● this work, – correlation, cf. equation (12), - - - critical temperature.

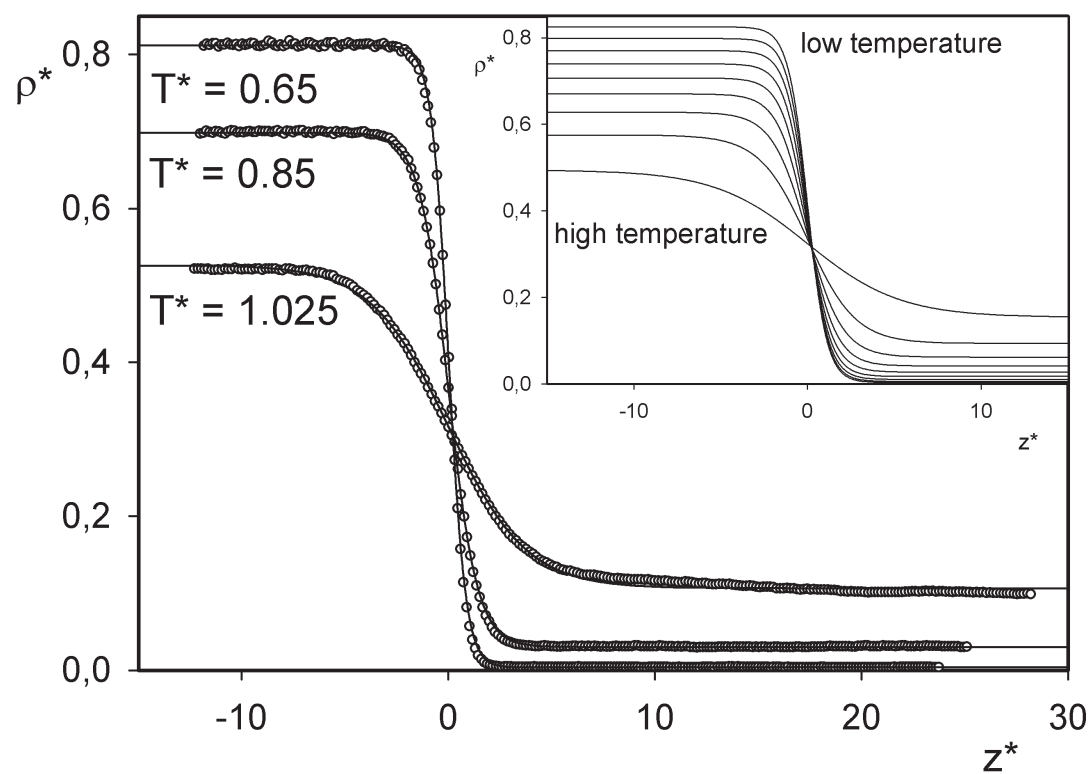


Figure 8. Large plot: density profiles of the planar interface for three different temperatures: $\circ$ simulation results, $-$ correlation, cf. equations (7), (8), and (12). Small plot: same correlation over the whole simulated temperature range $T^* = 0.625$ to 1.05 .

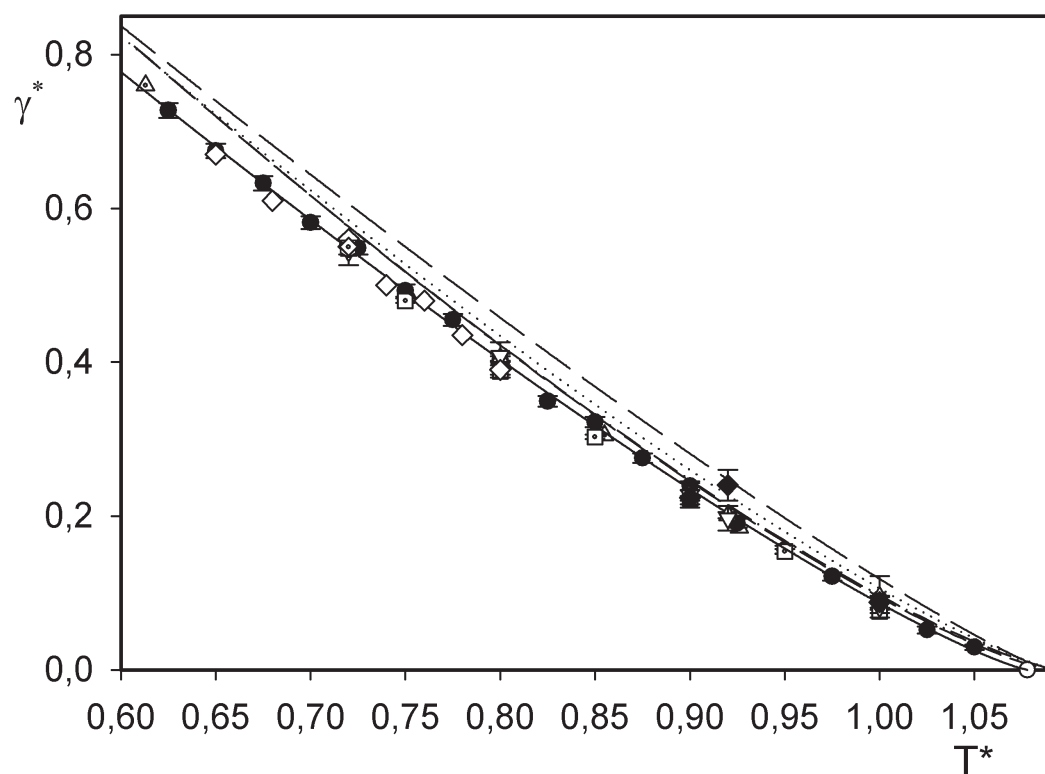


Figure 9. Surface tension vs. temperature for the truncated and shifted Lennard-Jones fluid: ● results from planar interface VLE simulations (this work), ○ critical point, – correlation, cf. equation (15), simulation results of other authors: □ Haye et al. [10], ◇ Mareschal et al. [5], ◇ Holcomb et al. [15], ○ Chen [14], △ Trokhymchuk and Alejandre (Monte-Carlo) [12], ▽ Trokhymchuk and Alejandre (molecular dynamics) [12], ◆ Nijmeijer et al. [13], ‘dotted triangle up’ Dunikov et al. [11]. Experimental real substance data reduced with the potential parameters from table 3: – · – Argon [40], · · · Krypton [41], - - - Xenon [41], - - Methane [42].

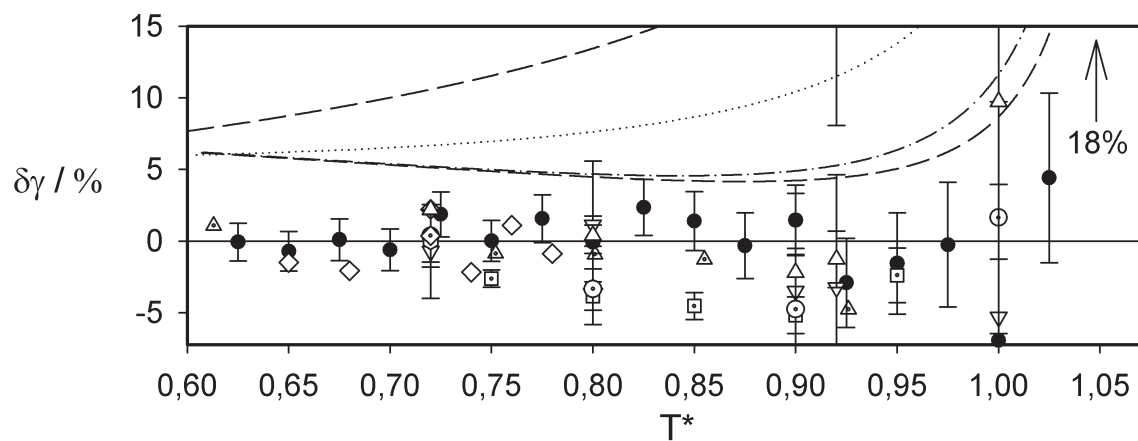


Figure 10. Deviation plot for the surface tension on the basis of the present correlation, cf. equation (15): ● this work, ○ critical point, □

Haye et al. [10], ◇ Mareschal et al. [5], ◇ Holcomb et al. [15], ○ Chen [14], △ Trokhymchuk and Alejandre (Monte-Carlo) [12], ▽ Trokhymchuk and Alejandre (molecular dynamics) [12], ◆ Nijmeijer et al. [13], ‘dotted triangle up’ Dunikov et al. [11]. Experimental real substance data reduced with the potential parameters from table 3: - · - Argon [40], · · · Krypton [41], - - - Xenon [41], - - Methane [42].

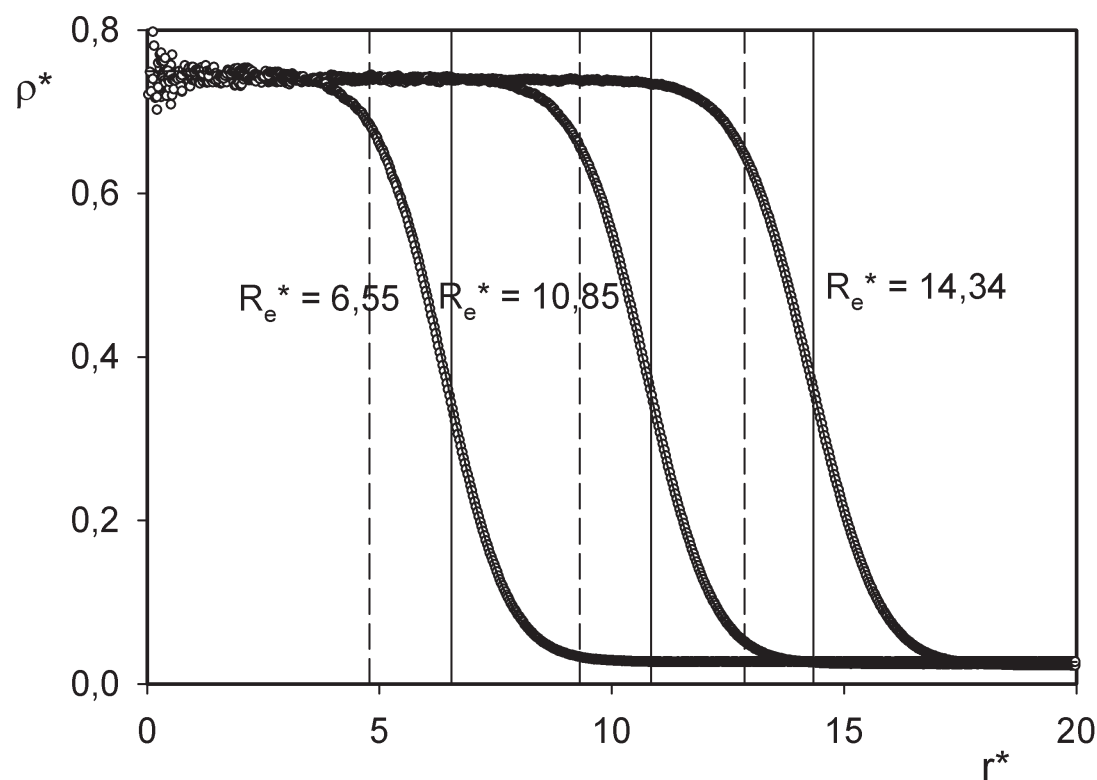


Figure 11. Density profiles of the spherical interface at $T^* = 0.80$ for three different droplet sizes: $\circ$ individual simulation results (this work),
 – generalized correlation, cf. equations (16) and (18) to (20). The solid vertical lines indicate R_e^* and the dashed vertical lines R_γ^* .

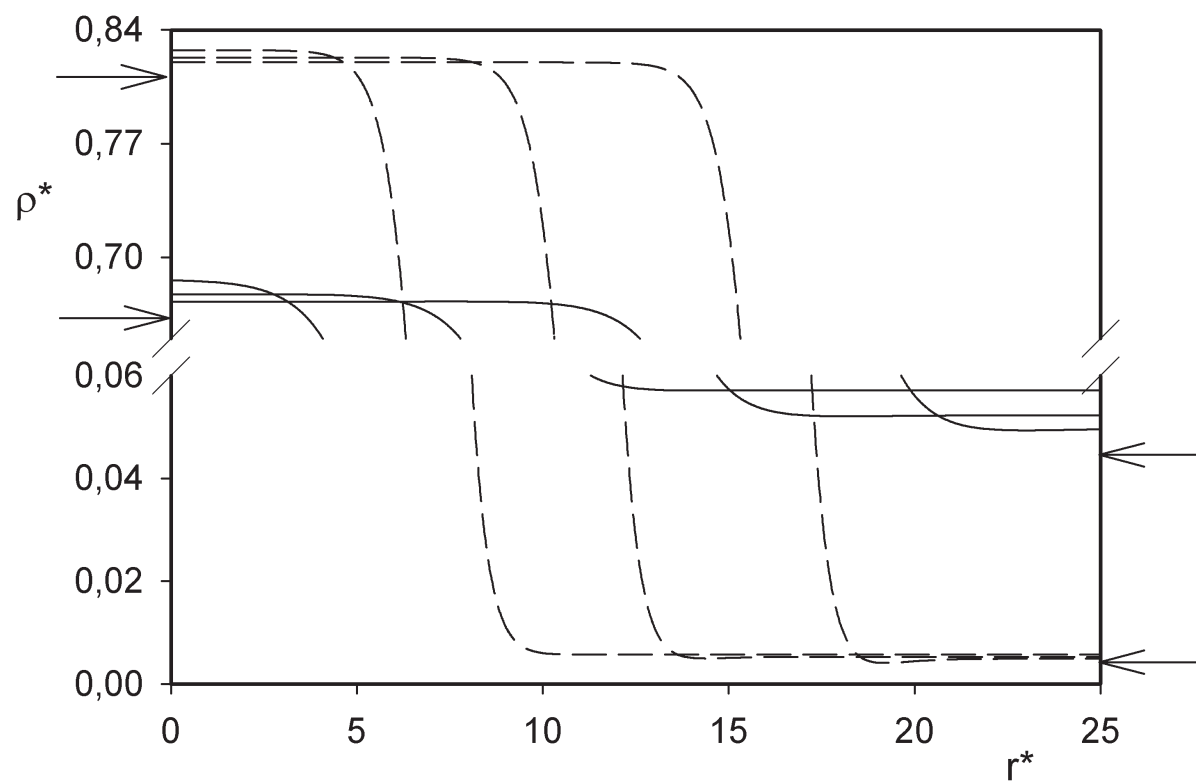


Figure 12. Density profiles of the spherical interface at $T^* = 0.65$ (—) and 0.90 (---) for three different droplet sizes $R_c^* = 7, 11$, and 16 from generalized correlation, cf. equations (16) and (18) to (20). The arrows indicate the homogeneous saturated densities.

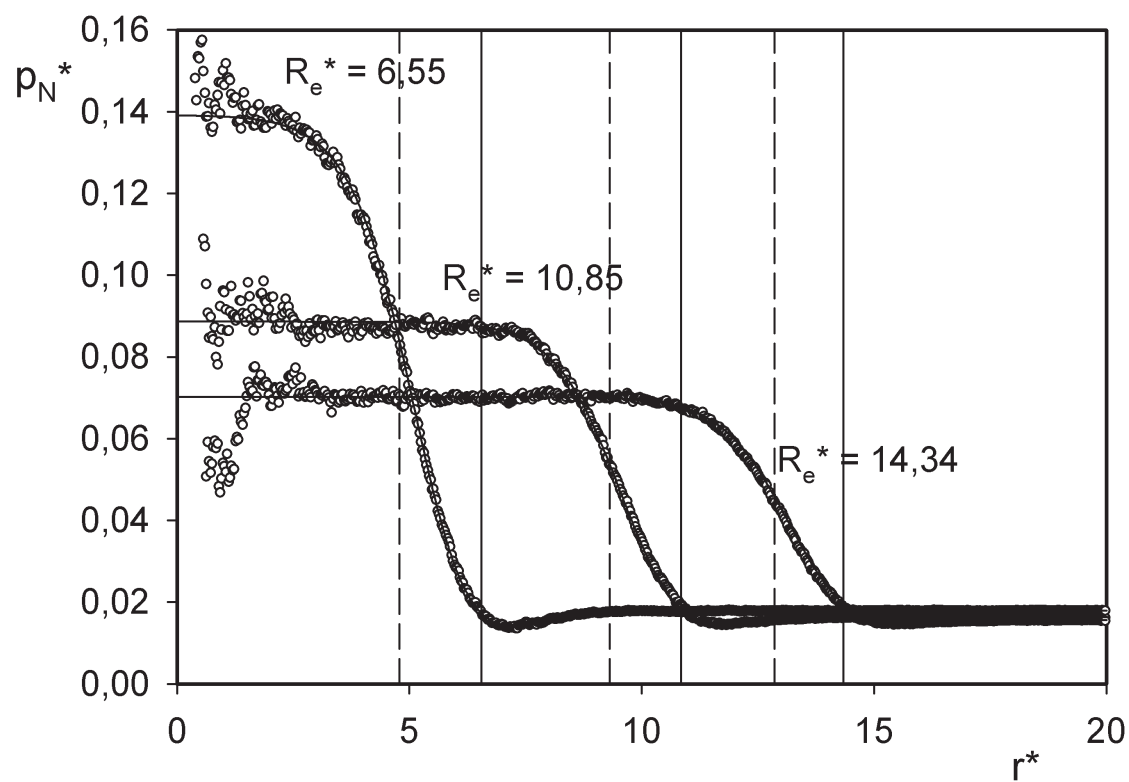


Figure 13. Normal pressure profiles of the spherical interface at $T^* = 0.80$ for three different droplet sizes: $\circ$ individual simulation results (this work), – generalized correlation, cf. equations (20) and (22) to (25). The solid vertical lines indicate R_e^* and the dashed vertical lines R_γ^* .

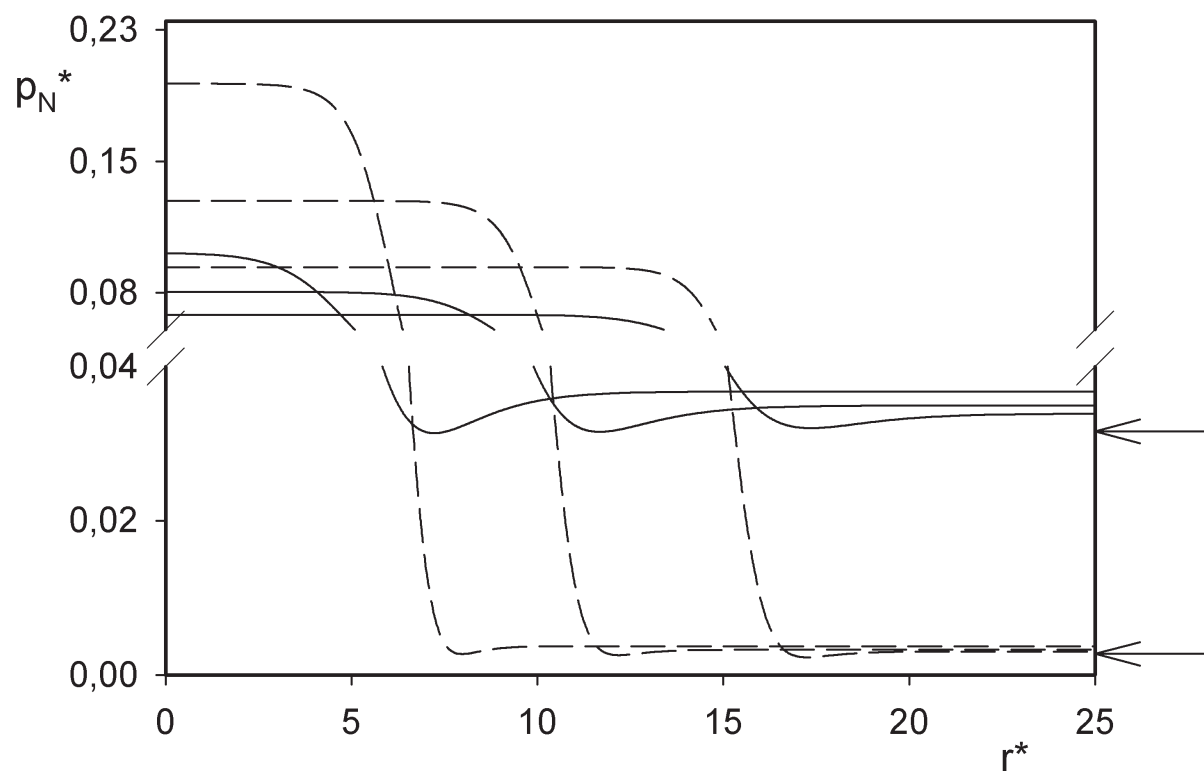


Figure 14. Normal pressure profiles of the spherical interface at $T^* = 0.65$ (—) and 0.90 (- - -) for three different droplet sizes $R_e^* = 7, 11$, and 16 from generalized correlation, cf. equations (20) and (22) to (25). The arrows on the right side indicate the homogeneous vapour pressure.

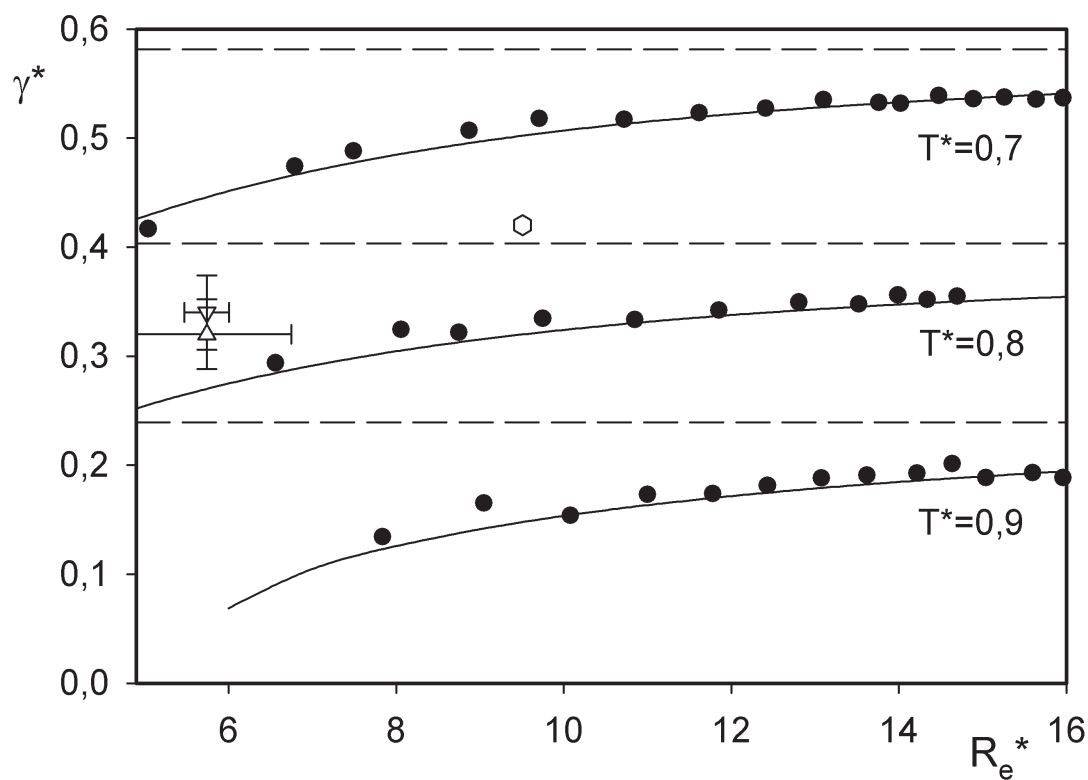


Figure 15. Surface tension vs. droplet size (i.e. Gibbs absorption dividing surface) for three different temperatures: ● individual simulation results (this work), — results from generalized correlation, cf. equations (20) and (22) to (26), -- results from planar interface VLE correlation, cf. equation (15), simulation results of other authors: Δ Thompson et al. (mechanical route) [45], ∇ Thompson et al. (thermodynamic route) [45], $\circ$ Bardouni et al. [46].

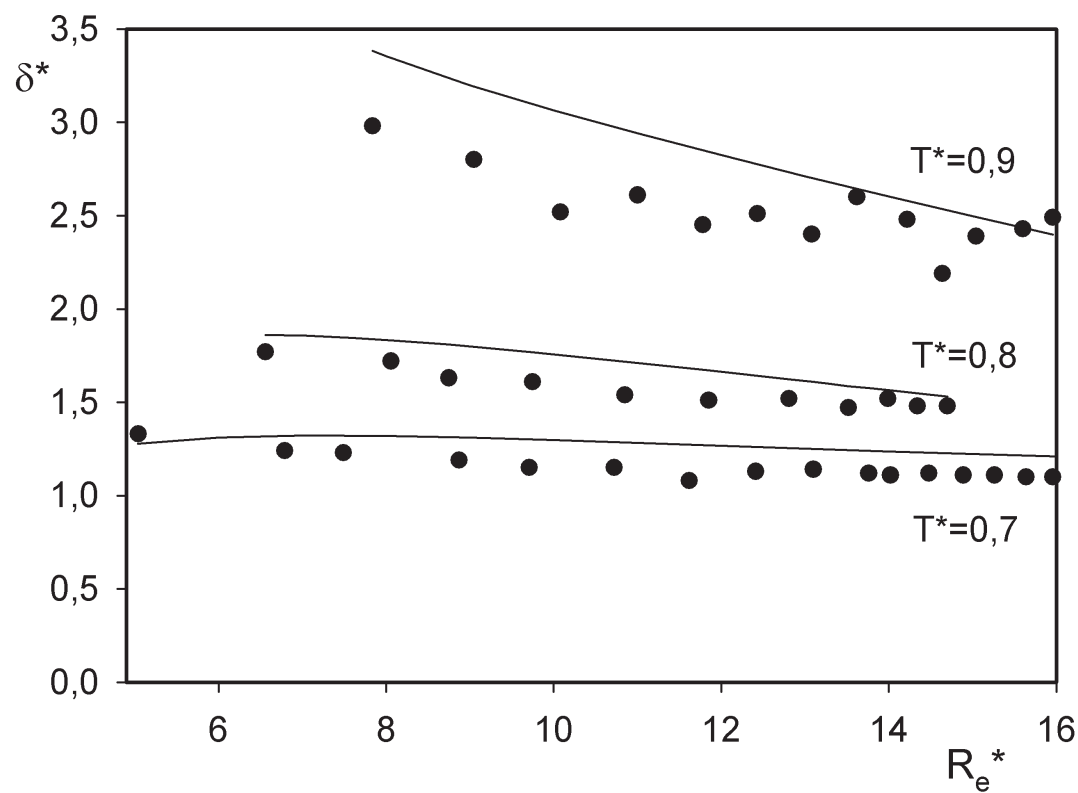


Figure 16. Tolman length vs. droplet size (i.e. Gibbs absorption dividing surface) of the spherical interface at $T^* = 0.80$: • individual simulation results (this work), – results from generalized density and normal pressure correlations.